Linear Algebra

Labix

June 12, 2026

Contents

1 Vector Spaces

1.1 Introduction to Vector Spaces

Definition 1.1.1 (Vector Space)
Let $\mathbb{F}$ be a field. A vector space over $\mathbb{F}$ is an abelian group V together with a function $\cdot : \mathbb{F} \times V \to V$ such that the following are true.

- $\lambda \cdot (v + w) = \lambda \cdot v + \lambda \cdot w$ for all $v, w \in V$ and $\lambda \in \mathbb{F}$
- $(\lambda\mu) \cdot v = \lambda \cdot (\mu \cdot v)$ for all $v \in V$ and $\lambda, \mu \in \mathbb{F}$
- $(\lambda + \mu) \cdot v = \lambda \cdot v + \mu \cdot v$ for all $v \in V$ and $\lambda, \mu \in \mathbb{F}$
- $1_{\mathbb{F}} \cdot v = v$ for all $v \in V$.

Example 1.1.2
Let $n \in \mathbb{N}^+$. Let $\mathbb{F}$ be a field. Then the following are true.

- $\mathbb{F}^n$ is a vector space over $\mathbb{F}$.
- $\mathbb{F}[x]$ is a vector space over $\mathbb{F}$.

Example 1.1.3
The set of all sequences

$$\mathbb{R}^{\mathbb{N}} = (a_n)_{n \in \mathbb{N}}$$

is a vector space.

Lemma 1.1.4
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $v \in V$. Let $\lambda \in \mathbb{F}$. Then the following are true.

- $\lambda \cdot 0 = 0$.
- $0 \cdot v = 0$.
- $(-\lambda) \cdot v = -(\lambda \cdot v) = \lambda \cdot (-v)$.
- If $\lambda \cdot v = 0$ then $\lambda = 0$ or $v = 0$.

Proof
We have that $\lambda \cdot 0 = \lambda \cdot (v - v) = \lambda \cdot v - \lambda \cdot v = 0$.

We have that $(\lambda - \lambda) \cdot v = \lambda \cdot v - \lambda \cdot v = 0$.

We have that $(-\lambda) \cdot v = ((-1)\lambda) \cdot v = (-1) \cdot (\lambda \cdot v)$ and $(-1)\lambda \cdot v = (\lambda(-1)) \cdot v = \lambda \cdot (-v)$.

If $\lambda \neq 0$, then $v = 1 \cdot v = \frac{1}{\lambda} \cdot (\lambda \cdot v) = \frac{1}{\lambda} \cdot 0 = 0$. ∎

1.2 Basis and Dimension

Definition 1.2.1 (Linearly Independent Vectors)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $S \subseteq V$ be a set of vectors. We say that the vectors in S are linearly independent if for any $v_1, \dots, v_n \in S$ and $a_1, \dots, a_n \in \mathbb{F}$ such that

$$\sum_{i=1}^{n} a_i v_i = 0$$

then we have $a_1 = \cdots = a_n = 0$.

Definition 1.2.2 (Span)

Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $S \subseteq V$. Define the span of S to be the set

$$\text{Span}(S) = \left\{ \sum_{i=1}^{n} a_i v_i \;\middle|\; a_1, \dots, a_n \in \mathbb{F}, v_1, \dots, v_n \in S \right\}$$

We also write $\mathbb{F}\langle v_1, \dots, v_n \rangle$ if we want to mention the ground field. Notice that the vectors in the span are a finite linear combination of the elements in the set. We may express the span of a subset $S \subseteq V$ of a vector space by the following notation:

$$\text{Span}(S) = \left\{ \sum_{s \in S} a_s s \;\middle|\; \text{only finitely many } a_s \in \mathbb{F} \text{ are non-zero} \right\}$$

Theorem 1.2.3 (Steinitz Exchange Lemma)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $U, W \subseteq V$ be finite subsets. If U is a set of linearly independent vectors and $\text{Span}(W) = V$, then the following are true.

- $|U| \leq |W|$
- There exists a set $W' \subset W$ with $|W'| = |W| - |U|$ such that $U \cup W'$ spans V.

Proof
Take $U = \{u_1, \dots, u_m\}$ and $W = \{w_1, \dots, w_n\}$. We will show that after reordering elements of W, we will have a set $\{u_1, \dots, u_m, w_m + 1, \dots, w_n\}$ that it spans V. We proceed by induction on m. Suppose that $m = 0$. In this case, $|U| \leq |W|$ necesssarily holds and by construction, W already spans V.
Now suppose that the proposition is true for $m - 1$. By the induction hypothesis, we may reorder elements of W so that $\{u_1, \dots, u_{m-1}, w_m, \dots, w_n\}$ spans V. Since $u_m \in V$, there exists $a_1, \dots, a_n \in \mathbb{F}$ such that

$$u_m = \sum_{k=1}^{m-1} a_k u_k + \sum_{k=m}^{n} a_k w_k$$

At least one of $a_m, \dots, a_n$ must be nonzero else the equality will contradict the linear independence of $u_1, \dots, u_m$. This must mean that $m \leq n$.
Now by reordering $a_m w_m, \dots, a_n w_n$, we may assume that $a_m \neq 0$. Thus we have that

$$w_m = \frac{1}{a_m} \left(u_m - \sum_{k=1}^{m-1} a_k u_k - \sum_{k=m+1}^{n} a_k w_k \right)$$

This means that w_m lies in the span of $\{u_1, \dots, u_m, w_{m+1}, \dots, w_n\}$. Since this span contains each of the vectors $u_1, \dots, u_{m-1}, w_m, \dots, w_n$, by the inductive hypothesis it spans V. ■

Clearly this implies that linearly independent sets of vectors must have cardinality less than sets of vectors that span V. By taking the highest cardinality of such linearly indendent set of vectors, and the lowest cardinality of such sets of vectors that span V, we necessarily have that they are equal and thus is exactly the dimension of V.

Definition 1.2.4 (Basis)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $S \subseteq V$. We say that S is a basis for V if the vectors in S are linearly independent and $\text{Span}(S) = V$.

The basis elements supporting a vector must be finite.

Example 1.2.5
Let $\mathbb{F}$ be a field. The standard basis vectors of $\mathbb{F}^n$ is denoted as $e_i = (0, \dots, 0, 1, 0, \dots, 0)$ where the one is in the ith slot, for $1 \leq i \leq n$.

Example 1.2.6
The set $\{(1,0,0,0,\dots),(0,1,0,0,\dots),(0,0,1,0,\dots),\dots\}$ is not a basis of $\mathbb{R}^{\mathbb{N}}$.

Proof
Indeed, the vector $(1,1,1,1,\dots)$ cannot be expressed as a finite linear combination of elements of the set. ■

Example 1.2.7
Let $\mathbb{F}$ be a field. Then the set $\{x^n \mid n \in \mathbb{N}\}$ is a basis for $\mathbb{F}[x]$.

Proposition 1.2.8
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Then V admits a basis.

We now give a criterion with matrices to find whether a set of vectors span V or whether they are linearly independent.

Proposition 1.2.9
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$ of dimension n. Let $B = \{b_1, \dots, b_n\}$ be a basis of V. Let $v_i = \sum_{j=1}^n a_{ij}b_j$ for $1 \leq i \leq m$ and some $a_{ij} \in \mathbb{F}$. Let $S = \{v_1, \dots, v_m\}$. Let

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}$$

Then the following are true.

- S is linearly independent if and only if the row echelon form of A has a leading one in every column
- S span V if and only if the row echelon form of A has no zero rows
- S is a basis of V if and only if the row echelon form of A is equal to the identity

Example 1.2.10
Let $B = \{b_1 = (1,1,0), b_2 = (1,0,1), b_3 = (0,1,1)\} \subseteq \mathbb{R}^3$. Then $S = \{v_1 = (1,1,2), v_2 = (1,2,1), v_3 = (2,1,1)\}$ is a basis for $\mathbb{R}^3$.

Proof
Notice that $v_1 = b_2 + b_3$, $v_2 = b_1 + b_3$ and $v_3 = b_1 + b_2$. We form the matrix

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

It is easy to check that $\text{rref}(A) = I_3$ so that S is a basis of $\mathbb{R}^3$. ■

Proposition 1.2.11
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $S, T \subseteq V$. If S and T are bases for V, then

$$|S| = |T|$$

Definition 1.2.12 (Dimension)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $S \subseteq V$ be a basis for V. Define the dimension of V to be

$$\dim(V) = |S|$$

We say that V is finite dimensional if $|S| < \infty$.

1.3 Vector Subspaces

Definition 1.3.1 (Vector Subspaces)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. A vector subspace of V is a subset $U \subseteq V$ such that U is a vector space over $\mathbb{F}$ in its own right.

Proposition 1.3.2 (Subspace Criterion)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $U \subseteq V$ be a subset. Then U is a subspace of V if and only if the following is true.

- $0_V \in U$
- For all $u_1, u_2 \in U$, $u_1 + u_2 \in U$
- For all $u \in U$ and $\lambda \in \mathbb{F}$, $\lambda \cdot u \in U$

Proof Suppose that U is a subspace of V. Then necessarily U is closed under vector addition and scalar multiplication and contains the zero vector.
Now suppose that the latter conditions are fulfilled by a subset U of V. Then it is easy to see that U satisfies all the criteria for being a vector space. ■

Proposition 1.3.3
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. If U_1 and U_2 are subspaces of V then $U_1 \cap U_2$ is also a subspace.

Proof Suppose that $\mathbf{v}, \mathbf{w} \in U_1 \cap U_2$. Then $\mathbf{v}, \mathbf{w} \in U_1$ and U_2. Since U_1, U_2 are subspaces, $\mathbf{v} + \mathbf{w} \in U_1$ and U_2 thus $\mathbf{v} + \mathbf{w} \in U_1 \cap U_2$. The proof is similar for scalar multiplication. ■

Definition 1.3.4 (Sum of Subspaces)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let U, W be subspaces of V. Then define the sum of U and W to be

$$U + W = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in U \text{ and } \mathbf{w} \in W\}$$

Proposition 1.3.5
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let U, W be subspaces of V. Then $U + W$ is the smallest subspace of V containing U and W.

Proof We first show that $U + W$ is indeed a subspace of V. Suppose that $v \in U + W$. Then there exists $u \in U$ and $w \in W$ such that $v = u + w$. Then since U and W are closed individually under vector addition and scalar multiplication, any product and addition in $U + W$ can be decomposed into a sum of vectors in U and W and thus the new vector will also be able to be decomposed into U and W and thus lie in $U + W$.

Now suppose that S is a subspace of V containing U and W. This means that any linear combination of elements of U and W are contained in S thus $U + W \subseteq S$. This means that if any subspace containing U and W must also contain $U + W$, which means that $U + W$ is the smallest subspace containing U and W. ■

Definition 1.3.6 (Independent Subspaces)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $W_1, \dots, W_n$ be subspaces of V. We say that $W_1, \dots, W_n$ are independent if for all $0 \neq w \in W_i$, we have $w \notin \sum_{j=1, j\neq i}^n W_j$.

Definition 1.3.7 (Direct Sum)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $W_1, \dots, W_n$ be subspaces of V. We say that V is the direct sum of $W_1, \dots, W_n$ if the following are true.

- $W_1, \dots, W_n$ are independent subspaces.
- $V = W_1 + \cdots + W_n$.

In this case, we write $V = W_1 \oplus \cdots \oplus W_n$.

Example 1.3.8
Let $e_1, \dots, e_4$ be the standard basis vectors of $\mathbb{R}^4$. Let $W_1 = \text{Span}(\{e_1 + e_2, e_3 - e_2\})$, $W_2 = \text{Span}(\{e_1 + e_3, e_4\})$. Notice that $e_1 + e_2, e_3 - e_2 \notin W_2$ and $e_1 + e_3, e_4 \notin W_1$, and they both have dimension 2. Hence $V = W_1 \oplus W_2$.

Corollary 1.3.9
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $W_1, \dots, W_n$ be subspaces of V. If $V = W_1 \oplus \cdots \oplus W_n$ then

$$\dim(V) = \sum_{k=1}^{n} \dim(W_k)$$

Proof Each basis of W_k are not contained in any other linear combination of all the basis of $W_1, \dots, W_{k-1}, W_{k+1}, \dots, W_n$. This means that the set of all the basis of $W_1, \dots, W_n$ are linearly independent. Since they each span W_k independently, the set of all the basis of $W_1, \dots, W_n$ will span $W_1 \oplus \cdots \oplus W_n$ and thus is a basis of V. Thus we are done. ■

1.4 Row and Column Ranks

This section is dedicated to the use of matrices and their relation to concepts in finite dimensional linear algebra.

Definition 1.4.1 (Row Space)
Let $\mathbb{F}$ be a field and let $A \in M_{m\times n}(\mathbb{F})$ be a matrix. The row space of A is the subspace of $\mathbb{F}^m$,

$$\text{Span}\{r_1, \dots, r_m\}$$

where r_i are the rows of A. The row rank of A is defined to be the dimension of the row space of A.

Definition 1.4.2 (Column Space)
Let $\mathbb{F}$ be a field and let $A \in M_{m\times n}(\mathbb{F})$ be a matrix. The column space of A is the subspace of $\mathbb{F}^n$,

$$\text{Span}\{c_1, \dots, c_n\}$$

where c_i are the columns of A. The column rank of A is defined to be the dimension of the column space of A.

Lemma 1.4.3
Let $\mathbb{F}$ be a field. Let $A \in M_{m\times n}(\mathbb{F})$ be a matrix. Let $B \in M_m(\mathbb{F})$ be a row operation matrix. Then A and BA have the same row space.

Proof
Suppose the rows of A are given by $r_1, \dots, r_n$. There are three cases to consider.

- Suppose that B is the scale matrix. Then there exists $\lambda \in \mathbb{F}$ such that the rows of BA are given by $r_1, \dots, r_{i-1}, \lambda r_i, r_{i+1}, \dots, r_n$. Clearly $\text{Span}(r_1, \dots, r_n) = \text{Span}(r_1, \dots, r_{i-1}, \lambda r_i, r_{i+1}, \dots, r_n)$.
- Suppose that B is the transposition matrix. Then clearly the rows of BA are also given by $r_1, \dots, r_n$ with possibly r_i and r_j swapped for some $i \neq j$. Then clearly they span the same space.
- Suppose that B is the recombine matrix. Then the rows of BA is given by $r_1, \dots, r_{i-1}, r_i + \lambda r_j, r_{i+1}, \dots, r_n$ for some $\lambda \in \mathbb{F} \setminus \{0\}$. Then clearly they also span the same space.

■

Proposition 1.4.4
The row rank of a matrix is equal to the column rank.

We can now define the rank of a matrix without problem.

Definition 1.4.5 (Rank of a Matrix)
Let $\mathbb{F}$ be a field. Let $A \in M_{m\times n}(\mathbb{F})$ be a matrix. Define the rank of A to be the row rank or column rank of A.

Proposition 1.4.6
Let $\mathbb{F}$ be a field. Let $A \in M_n(\mathbb{F})$ be a matrix. Then the following are equivalent.

- rank$(A) = n$.
- The rows of A form a linearly independent set.
- The columns of A form a linearly independent set.
- A is invertible.

Proof
$(1) \iff (2),(3)$: Suppose that the rank of A is n. Let $r_1,\dots,r_n$ be the rows of A. Then $r_1,\dots,r_n$ has spans a space of dimension n. Hence they are linearly independent. Now suppose that $r_1,\dots,r_n$ are linearly independent. Then they span a subspace of $\mathbb{R}^n$ of dimension n. Hence they span $\mathbb{R}^n$ and so $\text{rank}(A) = n$. The proof is similar for the column version.

$(1) \implies (4)$: Suppose that the rank of A is n. Since applying row operations do not change the rank, we know that the rows of rref(A) span a subspace of $\mathbb{R}^n$ of dimension n. By properties of the rref we conclude that rref$(A) = I$. Hence A is invertible.

$(4) \implies (1)$: Now suppose that A is invertible. Then the rref of A is I, and hence it has rank n. Since row operations do not change the rank, we conclude that rank$(A) = n$. ■

2 Preserving the Vector Space Structure

2.1 Linear Transformations

Definition 2.1.1 (Linear Transformation)
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Let $T : V \to W$ be a function. We say that T is a linear transformation if

$$T(\lambda_1 v_1 + \lambda_2 v_2) = \lambda_1 T(v_1) + \lambda_2 T(v_2)$$

for all $v_1, v_2 \in V$ and $\lambda_1, \lambda_2 \in \mathbb{F}$.

Equivalently, the conditions for a function to be a linear transformation can be expressed into:

- $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$.
- $T(\lambda v) = \lambda T(v)$ for all $v \in V$ and $\lambda \in \mathbb{F}$.

Example 2.1.2
Define the linear map $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T(x, y) = (2x - y, 3x + 2y)$. Then T is a linear map.

Example 2.1.3
Define the linear map $T : \mathbb{R}[x] \to \mathbb{R}[x]$ by $T(p(x)) = p'(x + 1)$. Then T is a linear map.

Lemma 2.1.4
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Let $T : V \to W$ be a linear map. Then the following are true.

- $T(0_v) = 0_w$
- $T(-v) = -T(v)$ for all $v \in V$

Proof Suppose that $v \in V$. Then $T(0 \cdot v) = 0 \cdot T(v) = 0$. Also we have that $T(0) = T(v + (-v)) = T(v) + T(-v)$. ■

Proposition 2.1.5
Let $\mathbb{F}$ be a field. Let U, V, W be vector spaces over $\mathbb{F}$. Let $T : V \to W$ and $S : W \to U$ be linear maps. Then $S \circ T$ is a linear map.

Proof
Let $au + bv \in V$. Then we have

$$S(T(au + bv)) = S(aT(u) + bT(v)) = a(S(T(u))) + b(S(T(v)))$$

■

Proposition 2.1.6 (Universal Property of Vector Spaces)
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Let B be a basis of V. Let $f : B \to W$ be a function. Then there exists a unique linear map $T : V \to W$ such that the following diagram commutes:

$$\begin{tikzcd} B \arrow[r, hook, "\iota"] \arrow[dr, "f"'] & V \arrow[d, dashed, "\exists! T"] \\ & W \end{tikzcd}$$

Proof
Let $v \in V$. Since B is a basis of V, there exists finitely many non-zero $\lambda_b \in \mathbb{F}$ such that $v = \sum_{b \in B} \lambda_b b$. Define the linear map $T : V \to W$ by

$$T(v) = \sum_{b \in B} \lambda_b f(b)$$

We check that this is a linear map. Let $v_1, v_2 \in V$. Let $a_1, a_2 \in \mathbb{F}$. Then $v_1 = \sum_{b\in B} \lambda_b b$ and $v_2 = \sum_{b\in B} \mu_b b$ for some $\lambda_b, \mu_b \in \mathbb{F}$. Then we have

$$\begin{aligned} T(a_1v_1 + a_2v_2) &= T\left(\sum_{b\in B}(a_1\lambda_b + a_2\mu_b)b\right) \\ &= \sum_{b\in B}(a_1\lambda_b + a_2\mu_b)f(b) \\ &= a_1\sum_{b\in B}\lambda_b f(b) + a_2\sum_{b\in B}\mu_b f(b) \\ &= a_1T(v_1) + a_2T(v_2) \end{aligned}$$

Hence T is linear.

It remains to show that T is unique. Suppose that L is another such linear map. Let $v = \sum_{b\in B} \lambda_b b \in V$. Then we have

$$T(v) = \sum_{b\in B}\lambda_b T(b) = \sum_{b\in B}\lambda_b f(b) = \sum_{b\in B}\lambda_b L(b) = L(v)$$

Hence $T = L$. ■

2.2 Isomorphisms

Definition 2.2.1 (Isomorphic Linear Maps)
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Let $T : V \to W$ be a linear map. We say that T is an isomorphism if there exists a linear map $S : W \to V$ such that

$$S \circ T = \mathrm{id}_V \quad \text{and} \quad T \circ S = \mathrm{id}_W$$

In this case we say that $V \cong W$.

Proposition 2.2.2
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Let $T : V \to W$ be a linear map. Then the following are equivalent.

- T is an isomorphism.
- T is bijective.
- $B \subseteq V$ is a basis for V if and only if $T(B)$ is a basis for W.

Proof
(1) $\implies$ (2): If T is an isomorphism then clearly T is bijective.

(2) $\implies$ (3): Let $B \subseteq V$ be a basis for V. Let $w \in W$ be a vector. Since T is surjective, there exists $v = \sum_{b\in B} \lambda_b b \in V$ such that $T(v) = w$. Since T is linear, we have $T(v) = \sum_{b\in B} \lambda_b T(b)$. Hence $T(B)$ spans W. Let $b_1, \dots, b_n \in B$. Suppose that $\sum_{i=1}^n a_i T(b_i) = 0$ for some $a_i \in \mathbb{F}$. Since T is linear, we have $T\left(\sum_{i=1}^n a_i b_i\right) = 0$. Since T is bijective and we know that $T(0) = 0$, we must have $\sum_{i=1}^n a_i b_i = 0$. But since B is a basis of V, we conclude that $a_i = 0$ for $1 \le i \le n$. Hence $T(B)$ is linearly independent. The other implication is done by the same argument using T^{-1}.

(3) $\implies$ (1): Let $B \subseteq V$ be a basis for V. Define a function $f : T(B) \to V$ as follows. Every element of $T(B)$ is of the form $T(b)$ for some $b \in B$. Then define $f(T(b)) = b$. By the universal property of vector spaces, there exists a unique linear map $S : W \to V$ such that $S(T(b)) =$

$f(T(b)) = b$. Let $v = \sum_{b \in B} \lambda_b b \in V$. Then we have

$$S(T(v)) = S\left(\sum_{b \in B} \lambda_b T(b)\right) = \sum_{b \in B} \lambda_b S(T(b)) = \sum_{b \in B} \lambda_b b = v$$

Hence $S \circ T = \text{id}_V$. Conversely, we know that $T(B)$ is a basis for W. Let $w = \sum_{b \in B} \lambda_b T(b) \in W$. Then we have

$$T(S(w)) = \sum_{b \in B} \lambda_b T(S(T(b))) = \sum_{b \in B} \lambda_b T(b) = w$$

since $S(T(b)) = b$. Hence $T \circ S = \text{id}_W$. Thus T is an isomorphism. ■

Corollary 2.2.3
Let $\mathbb{F}$ be a field. Let V be a finite dimensional vector space over $\mathbb{F}$. Then $V \cong \mathbb{F}^n$ for some $n \in \mathbb{N}$.

Proof
Choose a basis B of V. Suppose that $|B| = n$. Then choose a bijection from B to the standard basis vectors of $\mathbb{F}^n$. This extends to a linear map $V \to \mathbb{F}^n$ by the universal property. Since it is bijective on the basis, the above proposition shows that T is an isomorphism. ■

This corollary is especially important since it tells use that we only really need to study all of $\mathbb{R}^n$ to study all of finite dimensional spaces. Once we have our results on $\mathbb{R}^n$, we can translate it via an isomorphism.

Definition 2.2.4 (The Space of all Linear Maps)
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Define the space of all linear maps to be the set

$$\text{Hom}_{\mathbb{F}}(V, W) = \{T : V \to W \mid T \text{ is a linear map}\}$$

together with the following operations.
- For two linear maps $T, S : V \to W$, define its sum to be $T + S : V \to W$ defined by $(T + S)(v) = T(v) + S(v)$ for all $v \in V$
- For a linear map $T : V \to W$ and a scalar $\lambda \in \mathbb{F}$, define its product to be $\lambda T : V \to W$ defined by $(\lambda T)(v) = \lambda T(v)$ for all $v \in V$.

Lemma 2.2.5
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Then $\text{Hom}_{\mathbb{F}}(V, W)$ is a vector space over $\mathbb{F}$.

2.3 Kernels and Images

Definition 2.3.1 (Kernel of Linear Map)
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Let $T : V \to W$ be a linear map. Define the kernel of T to be the set

$$\ker(T) = \{v \in V \mid T(v) = 0\}$$

Since $T : V \to W$ can be regarded as a group homomorphism, the two notions of kernels coincide.

Proposition 2.3.2
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Let $T : V \to W$ be a linear map. Then the following are true.
- $\text{im}(T)$ is a subspace of W.
- $\ker(T)$ is a subspace of V.

Proof Clearly $0 \in \text{im}(T)$ since $T(0) = 0$. Let $w_1, w_2 \in \text{im}(T)$. Then there exists $v_1, v_2 \in V$ such that $T(v_1) = w_1$ and $T(v_2) = w_2$. Then $w_1 + w_2 = T(v_1) + T(v_2) = T(v_1 + v_2)$. Hence $w_1 + w_2 \in \text{im}(T)$. Let $w \in \text{im}(T)$. Let $\lambda \in \mathbb{F}$. Then there exists $v \in V$ such that $T(v) = w$. Then

$\lambda w = \lambda T(v) = T(\lambda v)$. Hence $\lambda w \in \operatorname{im}(T)$. Thus $\operatorname{im}(T)$ is a subspace of W by the subspace criterion.

Clearly $0 \in \ker(T)$ since $T(0) = 0$. Let $v_1, v_2 \in \ker(T)$. Let $\lambda_1, \lambda_2 \in \mathbb{F}$. Then we have

$$T(\lambda_1 v_1 + \lambda_2 v_2) = \lambda_1 T(v_1) + \lambda_2 T(v_2) = 0$$

Hence $\lambda_1 v_1 + \lambda_2 v_2 \in \ker(T)$. By the subspace criterion, we have that $\ker(T)$ is a subspace of V. ■

Definition 2.3.3 (Rank and Nullity)
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Let $T : V \to W$ be a linear map.

- Define the rank of T to be $\operatorname{rank}(T) = \dim(\operatorname{im}(T))$.
- Define the nullity of T to be $\operatorname{nullity}(T) = \dim(\ker(T))$.

Theorem 2.3.4 (Rank Nullity Theorem)
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$ such that V is finite dimensional. Let $T : V \to W$ be a linear map. Then we have

$$\operatorname{rank}(T) + \operatorname{nullity}(T) = \dim(V)$$

Proof
Suppose that $\dim(V) = n$. Let $v_1, \dots, v_k$ be a basis for $\ker(T)$. By the Steinitz exchange lemma, choose $w_1, \dots, w_{n-k} \in V$ such that $v_1, \dots, v_k, w_1, \dots, w_{n-k}$ is a basis for V. Now we know that

$$\operatorname{im}(T) = \operatorname{Span}\{T(v_1), \dots, T(v_k), T(w_1), \dots, T(w_{n-k})\} = \operatorname{Span}\{T(w_1), \dots, T(w_{n-k})\}$$

Hence $T(w_1), \dots, T(w_{n-k})$ spans $\operatorname{im}(T)$. It remains to show that they are linearly independent. Suppose for a contradiction that $\sum_{i=1}^{n-k} a_i T(w_i) = 0$ for at least one $a_i \in \mathbb{F}$ non-zero. Since T is linear, we have $T\left(\sum_{i=1}^{n-k} a_i w_i\right) = 0$. Hence $\sum_{i=1}^{n-k} a_i w_i \in \ker(T) = \operatorname{Span}(v_1, \dots, v_k)$. Hence there exists $b_i \in \mathbb{F}$ such that $\sum_{i=1}^{n-k} a_i w_i - \sum_{j=1}^{k} b_i v_i = 0$. Since there exists at least one a_i non-zero, this is a contradiction since $v_1, \dots, v_k, w_1, \dots, w_{n-k}$ is a basis by assumption and hence should be linearly independent. Thus $T(w_1), \dots, T(w_{n-k})$ is linearly independent. ■

2.4 Representing Linear Maps with Matrices

Definition 2.4.1 (Matrix of a Linear Map)
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Suppose that $\dim(V) = n$ and $\dim(W) = m$. Let $E = \{e_1, \dots, e_n\}$ be a basis of V. Let $F = \{f_1, \dots, f_m\}$ be a basis of W. Let $T : V \to W$ be a linear map. For $1 \leq i \leq n$, there exists $a_{ki} \in \mathbb{F}$ such that $T(e_i) = \sum_{k=1}^{m} a_{ki} f_k$. Define the matrix representing T under the bases E and F to be

$$[T]_E^F = \begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{m1} & \alpha_{m2} & \cdots & \alpha_{mn} \end{pmatrix} \in M_{m\times n}(\mathbb{F})$$

Example 2.4.2
Define the map $\frac{d}{dx} : \mathbb{R}[x]_{\leq 3} \to \mathbb{R}[x]_{\leq 3}$ on the basis $E = \{1, x, x^2, x^3\}$ by $\frac{d}{dx} x^n = n x^{n-1}$ for $0 \leq n \leq 3$. Then the matrix representing the linear map under the basis is given by

$$\left[\frac{d}{dx}\right]_E^E = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The same linear map under the basis $F = \{1, x+1, x^2+1, x^3+1\}$ is represented by

$$\left[\frac{d}{dx}\right]_F^F = \begin{pmatrix} 0 & 1 & -2 & -3 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Lemma 2.4.3
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Suppose that $\dim(V) = n$ and $\dim(W) = m$. Let E, F be bases for V, W respectively. Let $T : V \to W$ be a linear map. Then we have

$$T(v) = [T]_E^F v$$

for all $v \in V$.

Proposition 2.4.4
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Suppose that $\dim(V) = n$ and $\dim(W) = m$. Let E, F be bases for V, W respectively. Let $T : V \to W$ be a linear map. Then we have

$$\text{rank}(T) = \text{rank}([T]_E^F)$$

Proposition 2.4.5
Let $\mathbb{F}$ be a field. Let V, W, U be vector spaces over $\mathbb{F}$. Suppose that $\dim(V) = n$, $\dim(W) = m$ and $\dim(U) = k$. Let E, F, G be bases for V, W, U respectively. Let $T : V \to W$ and $S : W \to U$ be linear maps. Then we have

$$[S \circ T]_E^G = [S]_F^G [T]_E^F$$

Proposition 2.4.6
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Suppose that $\dim(V) = n$ and $\dim(W) = m$. Let E, F be bases for V, W respectively. Let $T : V \to W$ be a linear map. Then T is an isomorphism if and only if $[T]_E^F$ is invertible.

Proposition 2.4.7
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Suppose that $\dim(V) = n$ and $\dim(W) = m$. Let E, F be bases for V, W respectively. Then there is a vector space isomorphism

$$\text{Hom}_{\mathbb{F}}(V, W) \cong M_{m \times n}(\mathbb{F})$$

given by $T \mapsto [T]_E^F$.

2.5 Change of Bases

We have seen that matrices represent a linear map after choosing bases for the domain and codomain vector spaces. However, when does two matrices represent the same linear map but in different bases?

Definition 2.5.1 (The Change of Basis Matrix)
Let $\mathbb{F}$ be a field. Let V be a finite dimensional vector space over $\mathbb{F}$. Let $E = \{e_1, \dots, e_n\}$ and $F = \{f_1, \dots, f_n\}$ be bases for V. Define the change of basis map $C_E^F : V \to V$ to be the matrix representing the linear map $T : V \to V$ defined by $T(e_k) = f_k$ for $1 \leq k \leq n$.

Lemma 2.5.2
Let $\mathbb{F}$ be a field. Let V be a finite dimensional vector space over $\mathbb{F}$. Let E, F be bases for V. Then C_E^F is invertible.

Proposition 2.5.3
Let $\mathbb{F}$ be a field. Let V, W be finite dimensional vector spaces over $\mathbb{F}$. Let E, E' be bases for V. Let F, F' be bases for W. Let $T : V \to W$ be a linear map. Then we have

$$[T]_{E'}^{F'} = C_F^{F'}[T]_E^F C_{E'}^E$$

It is often easier to depict this as a diagram. Recall that we write V_B to indicate the vector space V is given the basis B. Using this notation we obtain the following commutative diagram as a result of the above theorem.

$$\begin{tikzcd} V_E \arrow[r, "{[T]_E^F}"] \arrow[d, "C_E^{E'}"'] & W_F \arrow[d, "C_F^{F'}"] \\ V_{E'} \arrow[r, "{[T]_{E'}^{F'}}"'] & W_{F'} \end{tikzcd}$$

Even better, we could have equivalently stated in the theorem that this commutative diagram is the result.

Definition 2.5.4 (Similar Matrices)
Let $\mathbb{F}$ be a field. We say that two matrices $A, B \in M_{n\times n}(\mathbb{F})$ are similar if there exists an invertible matrix $P \in GL_n(\mathbb{F})$ such that $B = PAP^{-1}$.

2.6 The Dual Space

Definition 2.6.1 (Dual Space) Let V be a vector space over a field $\mathbb{F}$. Define the dual space V^* of V to be the vector space

$$V^* = \text{Hom}(V, \mathbb{F})$$

of all linear forms on V

Explicitly, the vector space operations are defined as follows.

- For two linear forms $f, g : V \to \mathbb{F}$, define $f + g : V \to \mathbb{F}$ by $(f + g)(v) = f(v) + g(v)$ for all $v \in V$
- For a linear form $f : V \to \mathbb{F}$ and a scalar $\lambda \in \mathbb{F}$, define $\lambda f : V \to \mathbb{F}$ by $(\lambda f)(v) = \lambda \cdot f(v)$ for all $v \in V$

Definition 2.6.2 (Dual of a Basis)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let B be a basis of V. Define the dual of B to be the set $B^* = \{\phi_b \mid b \in B\}$ where $\phi_b \in V^*$ is given by

$$\phi_b(v) = \begin{cases} 1 & \text{if } v = b \\ 0 & \text{otherwise} \end{cases}$$

Proposition 2.6.3
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Then the following are true.
- B^* is a linearly independent set.
- B^* is a basis for V^* if and only if V is finite dimensional.

Corollary 2.6.4
Let $\mathbb{F}$ be a field. Let V be a finite dimensional vector space over $\mathbb{F}$. Then we have $\dim_{\mathbb{F}}(V^*) = \dim_{\mathbb{F}}(V)$.

Definition 2.6.5 (Dual Map) Let V, W be vector spaces over a field. Let $T \in \text{Hom}(V, W)$. The dual map of T is the linear map $T^* \in \text{Hom}(W^*, V^*)$ defined by $T^*(\phi) = \phi \circ T$ for $\phi \in W^*$.

Proposition 2.6.6
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Let $T, S : V \to W$ be linear maps. Let $\lambda, \mu \in \mathbb{F}$. Then we have
$$(\lambda T + \mu S)^* = \lambda T^* + \mu S^*$$

Proposition 2.6.7
Let $\mathbb{F}$ be a field. Then the following are true.
- Let V, W, U be vector spaces over $\mathbb{F}$. Let $T : V \to W$ and $S : W \to U$ be linear maps. Then we have $(S \circ T)^* = T^* \circ S^*$.
- Let V be a vector space over $\mathbb{F}$. Then $(\mathrm{id}_V)^* = \mathrm{id}_{V^*}$.

Proposition 2.6.8
Let $\mathbb{F}$ be a field. Let V, W be finite dimensional vector spaces over $\mathbb{F}$. Let $T : V \to W$ be a linear map. Let B, C be bases of V and W respectively. Then we have
$$[T^*]_{C^*}^{B^*} = ([T]_B^C)^T$$

Definition 2.6.9 (The Double Dual)
Let $\mathbb{F}$ be a field. Let V be a vector space over a field $\mathbb{F}$. Define the double dual of V to be the vector space
$$V^{**} = \mathrm{Hom}(\mathrm{Hom}(V, \mathbb{F}), \mathbb{F})$$

Proposition 2.6.10 Let V be a vector space over a field k. Define the function $\mathrm{ev}_V : V \to V^{**}$ by
$$\mathrm{ev}(v) = \begin{pmatrix} E_v : V^* \to \mathbb{R} \\ f \mapsto f(v) \end{pmatrix}$$
Then ev is an injective linear transformation. Moreover ev_V is natural in the following sense. If W is another vector space over k and $\phi : V \to W$ is a linear map, then the following diagram commutes:
$$\begin{tikzcd} V \arrow[r, "\phi"] \arrow[d, "\mathrm{ev}_V"'] & W \arrow[d, "\mathrm{ev}_W"] \\ V^{**} \arrow[r, "\phi^{**}"'] & W^{**} \end{tikzcd}$$

Corollary 2.6.11 Let V be a vector space over a field. If V is finite dimensional , then the map
$$\mathrm{ev} : V \to V^{**}$$
is an isomorphism.

Proof
Follows from the above since ev is injective. ■

3 Eigenspaces

Eigenspaces are invariants of a linear map. Every vector in the eigenspace will only be scaled while maintaining its direction.

3.1 Invariant Subspaces

Definition 3.1.1 (Invariant Subspaces)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Let $U \subseteq V$ be a subspace. We say that U is invariant under T if $T(U) \subseteq U$.

Lemma 3.1.2
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Then $\ker(T)$ and $\operatorname{im}(T)$ are invariant subspaces.

3.2 Eigenvalues and Eigenvectors

Definition 3.2.1 (Eigenvalues and Eigenvectors)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. An eigenvector of T is a vector $v \in V$ such that

$$T(v) = \lambda v$$

for some scalar $\lambda \in \mathbb{F}$. In this case we say that λ is an eigenvalue of T.

By choosing a basis for V an representing the linear map T by a matrix A, we notice that the concept of eigenvalues and eigenvectors can be translated to matrices.

Notice that two eigenvectors can have the same eigenvalue. For instance, if v is an eigenvector of λ and a is a scalar, then av is also an eigenvector of λ because

$$T(av) = aT(v) = a(\lambda v) = \lambda(av)$$

In particular, the collection of all eigenvectors of an eigenvalue forms a vector subspace of V.

Definition 3.2.2 (Eigenspace)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Let $\lambda \in \mathbb{F}$ be an eigenvalue of T. Define the eigenspace of λ to be the vector subspace

$$E_\lambda = \{v \in V \mid T(v) = \lambda v\}$$

Lemma 3.2.3
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Let $\lambda \in \mathbb{F}$ be an eigenvalue of T. Then

$$E_\lambda = \ker(T - \lambda I)$$

Proof Let $v \in V$. We have that

$$\begin{aligned} T(v) = \lambda v &\iff T(v) - \lambda v = 0 \\ &\iff (T - \lambda I)(v) = 0 \\ &\iff v \in \ker(T - \lambda I) \end{aligned}$$

and so we conclude. ∎

Proposition 3.2.4
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Let $\lambda_1, \dots, \lambda_r$ be distinct eigenvalues of T. Let $v_1, \dots, v_r$ be its eigenvectors respectively. Then $v_1, \dots, v_r$ are linearly independent.

Proof
Suppose that $\sum_{i=1}^r a_i v_i = 0$ for some $a_i \in \mathbb{F}$. Then for all $k \in \mathbb{N}$, applying T^k to the equation gives $\sum_{i=1}^r a_i \lambda_i^k v_i = 0$. We may write these equations into

$$\begin{pmatrix} 1 & \cdots & 1 \\ \lambda_1 & \cdots & \lambda_r \\ \vdots & \ddots & \vdots \\ \lambda_1^{r-1} & \cdots & \lambda_r^{r-1} \end{pmatrix} \begin{pmatrix} a_1 v_1 \\ \vdots \\ a_r v_r \end{pmatrix} = 0$$

The matrix on the left is the transpose of the Vandermonde matrix seen in Groups and Rings. The determinant of the matrix is given by $\prod_{1 \leq i < j \leq r}(\lambda_j - \lambda_i)$. Since the $\lambda_1, \dots, \lambda_r$ are distinct, the matrix is invertible. Applying the inverse on both sides give

$$\begin{pmatrix} a_1 v_1 \\ \vdots \\ a_r v_r \end{pmatrix} = 0$$

so $a_1 = \dots = a_r = 0$. Hence $v_1, \dots, v_r$ are linearly independent. ■

As we can see, distinct eigenvalues are linearly independent. Considering the span of each eigenvectors, we can clearly see that each of their spans are independent.

Proposition 3.2.5
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Let $\lambda_1, \dots, \lambda_r$ be distinct eigenvalues of T. Then $E_{\lambda_1}, \dots, E_{\lambda_r}$ are independent vector subspaces.

Proof
Clear from the fact that the basis of eigenspaces of different eigenvalues are linearly independent. ■

Proposition 3.2.6
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Let λ be an eigenvector of V. Then E_λ is T-invariant.

The main result of this subsection, stated that eigenspaces remain invariant under the linear transformation. Clearly this depends on the linear transformation. We will also show that this fact is also unchanged when considering different basis for the linear transformation.

3.3 Diagonalization

We now show a very nice kind of matrices, diagonal matrices that will come into play with linear maps. Our goal is to attempt to classify, by similarity, of every matrix into a diagonal one. We will soon see that this is not possible, and thus giving the last section of these notes meaning.

Definition 3.3.1 (Diagonalizable Linear Maps)
Let $\mathbb{F}$ be a field. Let V be a finite dimensional vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. We say that T is diagonalizable if there exists a basis E of V such that $[T]_E^E$ is diagonal.

We can transfer this concept to matrices as well.

Definition 3.3.2 (Diagonalizable Matrices)
Let $\mathbb{F}$ be a field. Let $A \in M_n(\mathbb{F})$ be a matrix. We say that A is diagonalizable if there exists an invertible matrix $P \in GL_n(\mathbb{F})$ such that $P^{-1}AP$ is diagonal.

Lemma 3.3.3
Let $\mathbb{F}$ be a field. Let V be a finite dimensional vector space over $\mathbb{F}$. Let E be basis of V. Let $T : V \to V$ be a linear map. Then T is diagonalizable if and only if $[T]_E^E$ is diagonalizable.

The compact way of saying this is that T is diagonalizable if and only if its equivalence class of similar matrices contain a diagonal matrix.

Ideally, we would like to study linear maps using the diagonal matrices because they appear to be very simple in nature. However, not every equivalence class of similar matrices contain a diagonal matrix and so this method is not feasible for every linear map. However, we can still find conditions for linear maps that are diagonalizable. This is highly related to eigenvalues and eigenspaces.

Proposition 3.3.4
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$ of dimension n. Let $T : V \to V$ be a linear map. If T has n distinct eigenvalues, then T is diagonalizable.

Although not stated in the theorem, this does not mean that linear maps without n distinct eigenvalues are not diagonalizable. However by taking the contrapositive, we see that not every linear map is diagonalizable because clearly, not every linear map has n distinct eigenvalues.

Proposition 3.3.5
Let $\mathbb{F}$ be a field. Let V be a finite dimensional vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues of T. Then following are equivalent.

- T is diagonalizable.
- V has a basis consisting of eigenvectors of T.
- $V = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_r}$.
- $\dim(V) = \dim(E_{\lambda_1}) + \cdots + \dim(E_{\lambda_r})$.

Proof
(1) $\implies$ (2): Suppose that E is a basis of V such that

$$[T]_E^E = \begin{pmatrix} a_1 & & \\ & \ddots & \\ & & a_{\dim(V)} \end{pmatrix}$$

By definition of the matrix representation, we have $T(e_i) = a_i e_i$ for $e_i \in E$. Hence E is a basis consisting of eigenvectors.

(2) $\implies$ (3): The basis of eigenvectors can be reorganized according to their corresponding eigenvalues.

(3) $\implies$ (4): Trivial. ■

4 The Jordan Canonical Form

In the last sections, we looked into what kinds of matrices can have "nice" looking matrix under some basis. We now provide a less "nice" looking form of a similar matrix. However, every matrix can be reduced to this relatively "nice" looking form, as long as the field is algebraically closed. This form is called the Jordan Normal Form.

4.1 The Minimal Polynomial

Definition 4.1.1 (The Minimal Polynomial)
Let $\mathbb{F}$ be a field. Let $A \in M_n(\mathbb{F})$ be a matrix. A minimal polynomial of A is a polynomial $\mu_A \in \mathbb{F}[x]$ such that the following are true.

- μ_A is monic.
- μ_A is of least degree such that $\mu_A(A) = 0$.

Proposition 4.1.2
Let $\mathbb{F}$ be a field. Let $A \in M_n(\mathbb{F})$ be a matrix. Then the minimal polynomial μ_A exists and is unique.

Proof
We first show that there exists a polynomial $p \in \mathbb{F}[x]$ such that $p(A) = 0$. Note that $\{I, A, \dots, A^{n^2}\}$ is linerarly dependent in the vector space of $n \times n$ matrices. Thus there exists constant $c_0, \dots, c_{n^2}$ that are not all zero such that

$$c_0 I + \cdots + c_{n^2} A^{n^2} = \mathbf{0}_n$$

Thus $p(x) = c_0 + c_1 x + \cdots + c_{n^2} x^{n^2}$ is our desired polynomial.

Thus there must exists a polynomial p of least degree such that $p(A) = 0$. We may divide the polynomial by its leading coefficient to obtain a monic polynomial.

For uniqueness, suppose that $q \in \mathbb{F}[x]$ is a monic polynomial not equal to p such that $q(A) = 0$. Then $\deg(p - q) < \deg(p)$ since p and q are monic. Moreover, $(p - q)(A) = p(A) - q(A) = 0$. Thus $p - q$ is also a minimal polynomial. But this contradicts the minimality of p. Hence the minimal polynomial is unique. ∎

Lemma 4.1.3
Let $\mathbb{F}$ be a field. Let $A \in M_n(\mathbb{F})$ be a matrix. Let $p \in \mathbb{F}[x]$ be a polynomial such that $p(A) = 0$. Then $\mu_A \mid p$.

Proof
By the division algorithm in Groups and Rings, there exists unique $q, r \in \mathbb{F}[x]$ such that $p = q\mu_A + r$ and $\deg(r) < \deg(\mu_A)$ or $r = 0$. If $\deg(r) < \deg(\mu_A)$, then $p(A) = q(A)\mu_A(A) + r(A)$ implies that $r(A) = 0$. Then r contradicts the minimality of μ_A. Thus $r = 0$ and $\mu_A \mid p$. ∎

Proposition 4.1.4
Let $\mathbb{F}$ be a field. Let $A, B \in M_n(\mathbb{F})$ be a matrices. If A and B are similar, then $\mu_A = \mu_B$.

Proof
Let $f \in \mathbb{F}[x]$. I claim that $f(P^{-1}AP) = P^{-1}f(A)P$. Suppose that $f = \sum_{i=0}^n a_i x^i$. Then we have

$$\begin{aligned} f(P^{-1}AP) &= \sum_{i=0}^n a_i (P^{-1}AP)^i \\ &= \sum_{i=0}^n a_i P^{-1}A^iP \\ &= P^{-1}\left(\sum_{i=0}^n a_i A^i\right)P \\ &= P^{-1}f(A)P \end{aligned}$$

Since A and B are similar, there exists $P \in GL_n(\mathbb{F})$ such that $A = P^{-1}BP$. Then we have $\mu_A(B) = P^{-1}\mu_A(A)P = 0$. Similarly, we have $\mu_B(A) = 0$. By the above lemma, we must have $\mu_A \mid \mu_B$ and $\mu_B \mid \mu_A$. Hence $\mu_A = \mu_B$. ■

Proposition 4.1.5
Let $\mathbb{F}$ be a field. Let $A \in M_n(\mathbb{F})$ be a matrix. Suppose that $P \in GL_n(\mathbb{F})$ such that $P^{-1}AP = \text{diag}(d_1, \dots, d_n)$ is diagonal. Then we have

$$\mu_A(x) = \prod_{i=1}^n (x - d_i)$$

Proof
Since similar matrices have the same minimal polynomial, we may assume that our matrix is diagonal. Notice that for any diagonal matrix $D = \text{diag}(d_1, \dots, d_n)$ and $f \in \mathbb{F}[x]$, we have $f(D) = \text{diag}(f(d_1), \dots, f(d_n))$. Thus $f(D) = 0$ if and only if $f(d_1) = \cdots = f(d_n) = 0$. Then the monic polynomial of least degree vanishing $d_1, \dots, d_n$ is given by $\mu_D(x) = (x - d_1)\cdots(x - d_n)$. ■

We will later see that in fact, the above criterion is a neccessary and sufficient condition: A is diagonalizable if and only if the minimal polynomial is a product of distinct linear factors.

4.2 The Characteristic Polynomial

Definition 4.2.1 (Characteristic Polynomial)
Let $\mathbb{F}$ be a field. Let $A \in M_n(\mathbb{F})$ be a matrix. Define the characteristic polynomial $c_A(x) \in \mathbb{F}[x]$ of A to be

$$c_A(x) = \det(A - xI_n)$$

Lemma 4.2.2
Let $\mathbb{F}$ be a field. Let $A, B \in M_n(\mathbb{F})$ be matrices, if A and B are similar, then $c_A(x) = c_B(x)$.

Theorem 4.2.3 (Cayley-Hamilton)
Let $\mathbb{F}$ be a field. Let $A \in M_n(\mathbb{F})$ be a matrix. Then $c_A(A) = 0$.

Proof
Firstly note that if $P(x) = \sum_{i=1}^n P_i x^i$ and $Q(x) = \sum_{j=1}^m Q_j x^j$ are polynomials with matrix coefficients where the matrix is $n \times n$, and $R(x) = \sum_{k=1}^{n+m} R_k x^k$ is the product of the two polynomials with $R_k = \sum_{i+j=k} P_i Q_j$, then if M is a $n \times n$ matrix that commmutes with all of Q_j, then we have

$$R(M) = P(M)Q(M)$$

This can be seen by expanding the sums out.
Now take $Q(x) = A - xI$ and $P(x) = \text{adj}(Q)$. Then we have $P(x)Q(x) = \det(A - xI)I = c_A(x)I$ by property of the adjoint. And since A commutes with all coefficients of the polynomial of Q, we

have

$$c_A(A)I = P(A)Q(A) = P(A) \cdot 0 = 0$$

Thus $c_A(A) = 0$.

Corollary 4.2.4
Let $\mathbb{F}$ be a field. Let $A \in M_n(\mathbb{F})$ be a matrix. Then $\mu_A \mid c_A$ and $\deg(\mu_A) \leq n$.

Proof
Follows from the Cayley Hamilton theorem and 4.1.3.

Proposition 4.2.5
Let $\mathbb{F}$ be a field. Let V be a finite dimensional vector space over $\mathbb{F}$. Let E be a basis for V. Let $T : V \to V$ be a linear map. Then the following are equivalent.

- $\lambda \in \mathbb{F}$ is an eigenvector for T.
- $c_{[T]_E^E}(\lambda) = 0$.
- $\mu_{[T]_E^E}(\lambda) = 0$.

Proof
$(1) \implies (3)$: Let v be an eigenvector of the eigenvalue λ of A. Trivially $\mu_A(A)v = 0$. But also since

$$A^n v = \lambda^n v$$

we have $0 = \mu_A(A)v = \mu_A(\lambda)v$. Since v is nonzero we must have $\mu_A(\lambda) = 0$.

Lemma 4.2.6 Let $\mathbb{F}$ be a field. Let $n, m \in \mathbb{N} \setminus \{0\}$. Let $A \in M_n(\mathbb{F})$ and $B \in M_m(\mathbb{F})$. Then

$$c_M(x) = c_A(x)c_B(x)$$

and

$$\mu_M(x) = \gcd(\mu_A(x), \mu_B(x))$$

In general, to deduce the formula for the minimal polynomial, we follow three steps.
Step 1: Find out the eigenvalues of the matrix.
Step 2: List out the possibilities of the minimal degree. This is done using the fact that $\mu_A(\lambda) = 0$ and $\deg(\mu_A) \leq n$.
Step 3: Plug in the matrix to find out which polynomial has its root at A.
There is another method to find out the formula using the following lemma.

Lemma 4.2.7
Let $\mathbb{F}$ be a field. Let V be a finite dimensional vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Let $W_1, \dots, W_k$ be the complete list of distinct invariant subspaces of T. Let $\mu_i(x)$ be the minimal polynomial of $T|_{W_i}$. Then

$$\mu_T(x) = \operatorname{lcm}(\mu_1, \dots, \mu_k)$$

Using this, we derive a better algorithm to find the minimal polynomial:
Step 1: Take $v \neq 0$ an eigenvector and set $W = \operatorname{span}\{v, T(v), T^2(v), \dots\}$. Then W is invariant under T. Let d be the minimal positive integer such that $v, T(v), \dots, T^d(v)$ are linearly dependent. Then $v, T(v), \dots, T^{d-1}(v)$ are linearly independent. Then we know that $\mu_T(x)$ has degree larger than d since else $\mu_T(x)v$ will never be 0. Then there is a nontrivial linear dependency relation of the form

$$T^d(v) + c_{d-1}T^{d-1}(v) + \cdots + c_1 T(v) + c_0 v = 0$$

Step 2: Consider the polynomial

$$x^d + c_{d-1}x^{d-1} + \cdots + c_1 x + c_0$$

Then this is precisely the minimal polynomial.

4.3 The Trace of a Linear Map

Definition 4.3.1 (The Trace of a Matrix)
Let k be a field. Let $M = (m_{i,j}) \in M_n(k)$ be a matrix. Define the trace of M to be

$$\text{tr}(M) = \sum_{i=1}^{n} m_{i,i}$$

Proposition 4.3.2
Let k be a field. Let $n \in \mathbb{N}^+$. Let $M \in M_n(k)$ be a matrix. Then the following are true.
- The constant term of the characteristic polynomial c_M is $\det(M)$.
- The coefficient of x^{n-1} in the characteristic polynomial c_M is $\text{tr}(M)$.

Lemma 4.3.3
Let k be a field. Then the following are true regarding the trace map $\text{tr} : M_n(k) \to k$.
- $\text{tr} : M_{n\times n}(k) \to k$ is a linear map
- $\text{tr}(AB) = \text{tr}(BA)$ for any $A, B \in M_n(k)$.
- If $A, B \in M_n(k)$ are such that A and B are similar, then $\text{tr}(A) = \text{tr}(B)$.

Lemma 4.3.4
Let k be a field. Let V be a finite dimensional vector space over k. Let $T : V \to V$ be a linear map. Let B, C be bases of V. Then we have

$$\text{tr}([T]_B^B) = \text{tr}([T]_C^C)$$

Definition 4.3.5 (The Trace of a Linear Map)
Let k be a field. Let V be a finite dimensional vector space over k. Let $T : V \to V$ be a linear map. Define the trace of T to be

$$\text{tr}(T) = \text{tr}([T]_B^B)$$

where B is any choice of basis of V.

Lemma 4.3.6
Let V be a vector space over $\mathbb{C}$. Let $T : V \to V$ be a linear map. Then we have

$$\text{tr}(T) = \sum_{\lambda \text{ is an eigenvalue of } T} \lambda$$

4.4 Generalized Eigenspace

Definition 4.4.1 (Generalized Eigenspaces)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Let $\lambda \in \mathbb{F}$ be an eigenvalue of T. Let $k \in \mathbb{N}^+$. Define the generalized eigenspace of index k of T with respect to λ to be

$$N_k(T, \lambda) = \ker((T - \lambda I)^k)$$

Elements of the subspace are called generalized eigenvectors.

Clearly we have an ascending chain of vector subspaces

$$N_1(T, \lambda) \subseteq N_2(T, \lambda) \subseteq \cdots$$

Proposition 4.4.2
Let $\mathbb{F}$ be a field. Let V be a finite dimensional vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map.

Let $\lambda \in \mathbb{F}$ be an eigenvalue of T. Then there exists $k \in \mathbb{N}^+$ such that

$$N_k(T,\lambda) = N_{k+n}(T,\lambda)$$

for all $n \in \mathbb{N}$.

Proof
Let $v \in N_i(T,\lambda)$. This means that $(T - \lambda \mathrm{id}_V)^i v = 0$. Applying the linear map $T - \lambda \mathrm{id}_V$ on both sides show that $v \in N_{i+1}(T,\lambda)$. ■

Definition 4.4.3 (Degree of Eigenvalues)
Let $\mathbb{F}$ be a field. Let V be a finite dimensional vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Let $\lambda \in \mathbb{F}$ be an eigenvalue of T. Define the degree of λ to be

$$d(\lambda) = \min\{k \mid N_k(T,\lambda) = N_{k+1}(T,\lambda) = \cdots\}$$

Recall that eigenvectors corresponding to different eigenvalues are linearly independent. The following is the corresponding generalization.

Proposition 4.4.4
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Let $\lambda_1, \dots, \lambda_r \in \mathbb{F}$ be eigenvalues of T. Let $v_i \in N_{d(\lambda_i)}(T,\lambda_i)$ for $1 \leq i \leq r$. Then $v_1, \dots, v_r$ are linearly independent.

Using generalized eigenspaces, we obtain an improvement of thm3.2.5.

Theorem 4.4.5 (Primary Decomposition Theorem)
Let $\mathbb{F}$ be an algebraically closed field. Let V be a vector space over $\mathbb{F}$. Let $T : V \to V$. Let $\lambda_1, \dots, \lambda_m$ be the full list of distinct eigenvalues of T. Then the following are true.

- V admits a decomposition into generalized eigenspaces

$$V = \bigoplus_{i=1}^m N_{d(\lambda_i)}(T,\lambda_i)$$

- The generalized eigenspace $N_{d(\lambda_i)}(T,\lambda_i)$ is an invariant subspace of T.

This version of the decomposition is applicable to any in vector space and linear map. In particular, if $d(\lambda_i) = 1$ for all i, notice that generalized eigenspaces are simply eigenspaces so that the two decompositions coincide.

4.5 Jordan Canonical Form

Definition 4.5.1 (Jordan Chain)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Let $\lambda \in \mathbb{F}$ be an eigenvalue of T. A Jordan Chain of length $k \in \mathbb{N}^+$ with respect to T and λ is a sequence of nonzero vectors $v_1, \dots, v_k \in V$ such that the following are true.

- $T(v_1) = \lambda v_1$
- For $2 \leq i \leq k$, we have $T(v_i) = \lambda v_i + v_{i-1}$

Lemma 4.5.2
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $T : V \to V$ be a linear map. Let $\lambda \in \mathbb{F}$ be an eigenvalue of T. Let $v_1, \dots, v_k \in V$ be a Jordan chain of length k with respect to T and λ. Then the following are true.

- For $1 \leq i \leq k$, we have $v_i \in N_i(A,\lambda)$
- For $2 \leq i \leq k$, $(T - \lambda \mathrm{id}_V)(v_i) = v_{i-1}$
- The vectors $v_1, \dots, v_k$ are linearly independent.

Proof ■

Definition 4.5.3 (Jordan Block of Degree k) Let $\mathbb{F}$ be a field. Let $\lambda \in \mathbb{F}$. Define the Jordan block of degree k of λ to be the $k \times k$ matrix

$$\gamma_{ij} = \begin{cases} \lambda & \text{if } j = i \\ 1_{\mathbb{F}} & \text{if } j = i+1 \\ 0 & \text{otherwise} \end{cases}$$

The matrix is denoted by $J_{\lambda,k}$.

Definition 4.5.4 (Jordan Basis) Let V be a vector space over a field $\mathbb{F}$. Let $T \in \text{End}_{\mathbb{F}}(V)$. A Jordan basis of T is sequence of Jordan chains $v_1, \dots, v_{k_1}, w_1, \dots, w_{k_2}, \dots$ for T such that it is a basis for V.

Proposition 4.5.5 Let V be a vector space over a field $\mathbb{F}$. Let $T \in \text{End}_{\mathbb{F}}(V)$. Let $v_1, \dots, v_n$ be a Jordan basis for T. Then the matrix representing T under the Jordan basis is given by a direct sum of Jordan blocks.

Theorem 4.5.6 Let $\mathbb{F}$ be an algebraically closed field. Let $n \in \mathbb{N} \setminus \{0\}$. Let $A \in M_n(\mathbb{F})$ be a matrix. Then the following are true.

- Consider A as a linear transformation $\mathbb{F}^n \to \mathbb{F}^n$. Then a Jordan basis for A exists.
- A is similar to a direct sum of Jordan blocks.

Proof We will construct the basis with 3 methods. They each contribute to part of the Jordan Basis. Firstly, we will obtain a basis of a subspace. We induct on n. The case of $n = 1$ is trivial.
Now suppose $T : V \to V$ is a linear map with $\dim(V) = n$. We want to find a restriction of T that is an automorphism to apply the induction hypothesis. Fix λ to be one of the eigenvalues of T. This will be used throughout the entire proof. This λ is possible because the ground field is algebraically closed. Now I claim that $U = \text{im}(T - \lambda I)$ is invariant under T. If this is true, then $T|_U : U \to U$ can be used to apply induction hypothesis. So all we have to show is that U is T-invariant and that $\dim(U) < \dim(V)$. The second item must be true by the rank nullity theorem. There must be an eigenvector in $\ker(T - \lambda I)$. Thus $\ker(T - \lambda I) \geq 1$ which implies that $\dim(U) < n$ by the rank nullity theorem. Now we prove that U is invariant. Let $u \in U$, I show that $T(u) \in U$. If $u \in U$, then $u = (T - \lambda I)(v)$ for some $v \in V$, hence

$$T(u) = T(T - \lambda I)(v) = (T - \lambda I)(T(v)) \in \text{im}(T - \lambda I) = U$$

Thus we have proven that $T|_U : U \to U$. Apply induction hypothesis here to obtain a Jordan Basis for T_U. Call that Jordan Basis $e_1, \dots, e_m$.
We now construct our second set of vectors. Recall that a Jordan Basis is a string of l Jordan Chains. Let $v_1, \dots, v_k$ denote one of the Jordan Chains. We can extend this Jordan Chain one more by setting $(T - \lambda I)^{k+1}(v_{k+1}) = 0$. Do the same thing for everey Jordan Chain, and relabel them to $w_1, \dots, w_l$. As a side note, the two set of vectors we have now still form a Jordan Basis because we simply extended every Jordan Chain one more.
For the final set of vectors, observe that the first vector of each of the l Jordan Chains are eigenvectors of $T|_U$ with its eigenvalue being λ. This is because by definition of Jordan Chains, $T|_U(v_1) = \lambda v_1$. Also note that those l vectors are linearly independent. Thus the first vectors of each of the l Jordan Chains span an l dimensional subspace of the eigenspace of λ. Recall that the eigenspace of λ has dimension $\dim(V) - \dim(U) = \dim(\ker(T - \lambda I))$. To minimize notation let $m = \dim(U)$. Thus by extension theorem we can extend the basis of the l dimensional subspace to $\ker(T - \lambda I)$. Call the extension vectors $w_{l+1}, \dots, w_{n-m}$. As a side note, these $n - m - l$ vectors each Jordan Chains of length 1. Thus we have complete our last set of vectors.
We have n vectors

$$e_1, \dots, e_m, w_1, \dots, w_l, w_{l+1}, \dots, w_{n-m}$$

Thus we only need to prove that they are linearly independent. Let $x = \sum_{k=1}^{m} \beta_m e_m$. Let

$$\sum_{i=1}^{n-m} \alpha_i w_i + x = 0$$

Applying $T - \lambda I$ on both sides give

$$\sum_{i=1}^{l} \alpha_i (T - \lambda I) w_i + (T - \lambda I)(x) = 0$$

Since $w_{l+1}, \dots, w_{n-m}$ is in $\ker(T - \lambda I)$, they become 0. Now recall that our construction of $w_1, \dots, w_l$ is made by extending our Jordan Chains. So applying $(T - \lambda I)$ moves down our Jordan Chain. This means that $(T - \lambda I)x$ no longer contains the last term of each Jordan Chain and are linear combinations of $\{e_1, \dots, e_m\} \setminus \{\text{Last Term of each Jordan Chain}\}$, while all of the $(T - \lambda I)(w_1), \dots, (T - \lambda I)(w_l)$ are all last members of ecah Jordan Chain. From the fact that $e_1, \dots, e_m$ are a basis, we have $\alpha_1 = \cdots = \alpha_l = 0$.
Our sum now becomes $(T - \lambda I)x = 0$. Which means that $x \in \ker(T - \lambda I)$. Now our original sum becomes

$$\sum_{i=l+1}^{n-m} \alpha_i w_i + x = 0$$

By construction, $w_{l+1}, \dots, w_{n-m}$ extends a basis of the eigenspace of $T|_U$ for λ, thus $\alpha_{l+1} = \cdots = \alpha_{n-m} = 0$. Also since $e_1, \dots, e_m$ is a basis of U, we have $\beta_1 = \cdots = \beta_m = 0$. Finally by the above corollary, in a Jordan Basis, the matrix of T is a direct sum of Jordan Blocks. ■

Definition 4.5.7 (The Jordan Canonical Form) Let $\mathbb{F}$ be an algebraically closed field. Let $n \in \mathbb{N} \setminus \{0\}$. Let $A \in M_n(\mathbb{F})$ be a matrix. Define the Jordan canonical form of A to be any choice of direct sum of Jordan blocks given by the above theorem.

Lemma 4.5.8 Let $\mathbb{F}$ be an algebraically closed field. Let $n \in \mathbb{N} \setminus \{0\}$. Let $A \in M_n(\mathbb{F})$ be a matrix. Then A and B have the same Jordan canonical form up to a reordering of the direct sum in the JCF.

Proposition 4.5.9 Let $\mathbb{F}$ be an algebraically closed field. Let $n \in \mathbb{N} \setminus \{0\}$. Let $A \in M_n(\mathbb{F})$ be a matrix. Then the following are true.

- The number of Jordan Blocks of J with eigenvalue λ is equal to $\dim(\ker(A - \lambda I))$
- The number of Jordan Blocks of J with eigenvalue λ of degree at least i is equal to $\dim(N_i(A, \lambda)) - \dim(N_{i-1}(A, \lambda))$

Proof Since similar matrices have the same dimensions for their generalized eigenspaces corresponding to their eigenvalue, WLOG take $A = J = J_{\lambda_1, k_1} \oplus \cdots \oplus J_{\lambda_s, k_s}$. However, note that the dimension of $N_i(A \oplus B, \lambda)$ is equal to $\dim(N_i(A, \lambda)) + \dim(N_i(B, \lambda))$. So we just have to prove the theorem for a single Jordan Block.
Since $(J_{\lambda,k} - \lambda I)^i$ has a single diagonal line of ones i places above the diagonal for $i < k$, and is 0 for $i \geq k$, the dimension of its kernel is i for $0 \leq i \leq k$ and k for $i \geq k$. ■

Proposition 4.5.10 Let $\mathbb{F}$ be an algebraically closed field. Let $n \in \mathbb{N} \setminus \{0\}$. Let $A \in M_n(\mathbb{F})$ be a matrix. Then the Jordan canonical form of A is unique up to reordering of the direct sum of the Jordan blocks.

Proposition 4.5.11 Let V be a vector space over an algebraically closed field $\mathbb{F}$. Let $T \in \text{End}_k(V)$. Let $\lambda_1, \dots, \lambda_m$ be the full list of distinct eigenvalues of T. Then then following are true.

- The characteristic polynomial of T is given by

$$c_T(x) = (-1)^n \prod_{k=1}^{m} (x - \lambda_k)^{a_k}$$

  where a_k is the sum of the degrees of the Jordan Blocks of T of eigenvalue λ_k

- The minimal polynomial of T is

$$\mu_A(x) = \prod_{k=1}^{m}(x-\lambda_k)^{d(\lambda_k)}$$

Proposition 4.5.12 Let V be a vector space over an algebraically closed field $\mathbb{F}$. Let $T \in \text{End}_k(V)$. Then the following are equivalent.
- T is diagonalizable.
- $\mu_T(x)$ has no repeated factors.
- $d(\lambda) = 1$ for all eigenvalues λ of T.

Proposition 4.5.13 Let $A, B \in M_{n\times n}(\mathbb{C})$. Then A and B are similar if and only if the following are true.
- A and B have the same set of eigenvalues
- $\dim(\ker(A-\lambda I)^i) = \dim(\ker(B-\lambda I)^i)$ for all i and eigenvalues λ

To finish this section, we show the process of determining the Jordan Canonical form of a matrix. The steps are usually as follows:
Step 1: Find out the nullity of $A-\lambda I$ as this gives us the number of Jordan Blocks with eigenvalue λ.
Step 2: To find out the number of Jordan blocks with eigenvalue λ and size at least i, we calculate $\dim(\ker(A-\lambda I_n)^i) - \dim(\ker(A-\lambda I)^{i-1})$.
To find out the change of basis matrix, meaning the Jordan Chains, we do the following step:
Step 3: Find out the last one in the chain v_k by solving $(A-\lambda I)^k v_k = 0$ while restricting v_k such that $(A-\lambda I)^{k-1}v_k \neq 0$, and then proceed to find out v_{k-1} by $(A-\lambda I)^{k-1}v_k = v_{k-1}$ and vice versa.

There are also extra information that we can use to determine the JCF:
The degree of $(x-\lambda)$ in μ_A indicates the maximum size of Jordan Blocks with eigenvalue λ.
The degree of $(x-\lambda)$ in c_A indicates the total size of used in the JCF of all Jordan Blocks with eigenvalue λ.

We give an example of finding the JCF of a matrix.

Example 4.5.14 Find the Jordan Canonical Form of

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}$$

Proof We begin by finding out the eigenvalues of A. We have that $c_A(x) = \det(A-xI) = (1-x)^2(2-x)$. This means that the eigenvalues are 1 and 2. Now we begin with step 1.
For eigenvalue 1, we have that row reduced form of $A-I$ is

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Thus the nullity of $A-I$ is 1. This means that the number of Jordan blocks with eigenvalue 1 is 1. Using information from c_A, we know that the total size used for the eigenvalue 1 is 2. This means that there is exactly one Jordan block of size 2 in the JCF of A.
This leaves the fact that the remaining Jordan block of size 1 being the eigenvalue 2.
With this, we complete the JCF of A with

$$J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

To compute the basis, or the change of basis matrix P, we use step 3. Since A has one Jordan block for eigenvalue 1, we need to find one string of Jordan chain. This chain needs to have length 2 since the size of the Jordan block is 2. (If there are multiple of Jordan blocks of the same eigenvalues, the end vector of the Jordan chains needs to be linearly indendent). We begin by finding the ending of the chain, v_2 by using the fact that $(A-I)^2v_2 = 0$ and $(A-I)v_2 = v_1 \neq 0$. We have that

$$(A-I)^2 = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

We choose that $v_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$. Now we have

$$v_1 = (A-I)v_2 = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$$

Finally, we choose $v_3 \in \ker(A-2I)$. But row reducing $A-2I$ gives

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

We can choose $v_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$. This means that

$$P = \begin{pmatrix} -1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

■

4.6 The Jordan-Chevalley Decompositions

Definition 4.6.1 (Jordan-Chevalley Decompositions) Let k be a field. Let V be a finite dimensional vector space over k. Let $T \in \text{End}(V)$. A Jordan-Chevalley decomposition of V consists of $D, S \in \text{End}(V)$ such that the following are true.

- $T = D + S$
- D is diagonalizable
- S is nilpotent
- $SD = DS$

We note here that if we consider vector spaces as a k-module, then saying V semisimple is the same as saying V is diagonalizable.

Proposition 4.6.2 Let k be a field. Let V be a finite dimensional vector space over k. Let $T \in \text{End}(V)$. Then T admits a unique Jordan-Chevalley decomposition.

Proof Let $T = PJP^{-1}$ be the Jordan canonical form of x. Then J has non-zero entries exactly at the (i,i)th and $(i,i+1)$ positions. Therefore we can decompose J as a sum of a diagonal matrix D_J and a matrix D_J with non-zero entry only on $(i,i+1)$th positions. But S_J must then be nilpotent. Since J is given in block form, we can also consider D_J and S_J as blocks. But diagonal matrices commute with all matrices. Hence $D_JS_J = S_JD_J$. Now let $D = PD_JP^{-1}$ and $S = PS_JP^{-1}$. I claim that D, S consist of ta Jordan-Chevalley decomposition. We have that $T = PJP^{-1} = P(D_J + S_J)P^{-1} = PD_JP^{-1} + PS_JP^{-1} = D + S$. Also, D is diagonalizable since $D = PD_JP^{-1}$ and D_J is diagonalizable. Now we know that $S_J^n = 0$ for some $n \in \mathbb{N}$. Then $S^n = PS_J^nP^{-1} = 0$ hence S is also nilpotent. Finally, we have that

$$DS = PD_JP^{-1}PS_JP^{-1} = PD_JS_JP^{-1} = PS_JD_JP^{-1} = SD$$

hence we proved the existence of such a decomposition. ■

Proposition 4.6.3 Let k be an algebraically closed field. Let V be a finite dimensional vector space over k. Let $T \in \text{End}(V)$. Let $D, S \in \text{End}(V)$ be the Jordan-Chevalley decomposition of T. Then there exists $p, q \in k[x]$ such that $p(T) = D$ and $q(T) = S$.

5 Linearity in Two Variables

5.1 Bilinear Maps

Definition 5.1.1 (Bilinear Maps)
Let $\mathbb{F}$ be a field. Let V, W be vector spaces over $\mathbb{F}$. Let $B : V \times W \to \mathbb{F}$ be a function. We say that B is a bilinear map if the following are true.

- $B(-, w) : V \to \mathbb{F}$ is a linear map for all $w \in W$.
- $B(v, -) : W \to \mathbb{F}$ is a linear map for all $v \in V$.

Definition 5.1.2 (Matrix Representing Bilinear Maps)
Let $\mathbb{F}$ be a field. Let V, W be finite dimensional vector spaces over $\mathbb{F}$. Let E be a basis of V and let F be a basis of W. Let $B : V \times W \to \mathbb{F}$ be a bilinear map. Define the matrix representing B under the basis E and F to be

$$[B]_{E,F} = \begin{pmatrix} B(e_1, f_1) & \cdots & B(e_1, f_m) \\ \vdots & \ddots & \vdots \\ B(e_n, f_1) & \cdots & B(e_n, f_m) \end{pmatrix}$$

Proposition 5.1.3
Let $\mathbb{F}$ be a field. Let V, W be finite dimensional vector spaces over $\mathbb{F}$. Let E be a basis of V and let F be a basis of W. Let $\tau : V \times W \to \mathbb{F}$ be a bilinear map. Then for any $v \in V$ and $w \in W$, we have

$$\tau(v, w) = v^T [\tau]_{E,F} w$$

Proof Simple to see by expanding the matrix multiplication. ■

Proposition 5.1.4
Let $\mathbb{F}$ be a field. Let V, W be finite dimensional vector spaces over $\mathbb{F}$. Let E, E' be bases of V and let F, F' be bases of W. Then we have

$$[\tau]_{E',F'} = (C_{E'}^{E})^T [\tau]_{E,F} C_{F'}^{F}$$

Proposition 5.1.5
Let $\mathbb{F}$ be a field. Let V, W be finite dimensional vector spaces over $\mathbb{F}$. Let E, F be bases of V and W respectively. Then there is a bijection

$$\{B : V \times W \to \mathbb{F} \mid B \text{ is bilinear } \} \overset{1:1}{\leftrightarrow} M_{\dim(W) \times \dim(V)}(\mathbb{F})$$

given by $B \mapsto [B]_{E,F}$.

Definition 5.1.6 (Perfect Pairings)
Let k be a field. Let V, W be vector spaces over k. Let $\tau : V \times W \to k$ be a bilinear map. We say that τ is a perfect pairing if the following are true.

- The map $V \to W^*$ given by $v \mapsto \tau(v, -)$ is an isomorphism.
- The map $W \to V^*$ given by $w \mapsto \tau(-, w)$ is an isomorphism.

Proposition 5.1.7
Let k be a field. Let V, W be finite dimensional vector spaces over k. Let $\tau : V \times W \to k$ be a bilinear map. Then the following are equivalent.

- τ is a perfect pairing.
- The map $V \to W^*$ given by $v \mapsto \tau(v, -)$ is an isomorphism.
- The map $W \to V^*$ given by $w \mapsto \tau(-, w)$ is an isomorphism.
- The matrix representing τ over any bases of V and W is invertible.

5.2 Bilinear Forms

Definition 5.2.1 (Bilinear Forms)
Let V be a vector space over a field $\mathbb{F}$. A bilinear form on V is a bilinear map

$$\tau : V \times V \to \mathbb{F}$$

A bilinear form naturally induces a comparison map $V \to V^*$ by $v \mapsto \tau(v, -)$.

Definition 5.2.2 (Symmetric and Skew-Symmetric of Bilinear Forms) Let k be a field. Let V be a vector space over k. Let $\tau : V \times V \to k$ be a bilinear form.

- We say that τ is symmetric if $\tau(v, w) = \tau(w, v)$.
- We say that τ is skew-symmetric if $\tau(v, w) = -\tau(w, v)$.
- We say that τ is alternating if $\tau(v, v) = 0$.

Lemma 5.2.3
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let τ be a bilinear form. Then the following are true.

- If τ is alternating, then τ is skew-symmetric.
- If $\text{char}(\mathbb{F}) \neq 2$, then τ is alternating if and only if τ is skew-symmetric.

Proposition 5.2.4 Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $\tau : V \times V \to \mathbb{F}$ be a bilinear form. Then the following are true.

- τ is symmetric if and only if the matrix representing τ over any basis is symmetric.
- τ is skew-symmetric if and only if the matrix representing τ over any basis is skew-symmetric.

Proposition 5.2.5
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $B : V \times V \to \mathbb{F}$ be a bilinear form. Then the following are true.

- The function $B_s : V \times V \to \mathbb{F}$ given by $B_s(u, v) = \frac{1}{2}(B(u, v) + B(v, u))$ is a symmetric bilinear map.
- The function $B_a : V \times V \to \mathbb{F}$ given by $B_a(u, v) = \frac{1}{2}(B(u, v) - B(v, u))$ is a skew-symmetric bilinear map.

Definition 5.2.6 (Non-Degenerate of Bilinear Forms)
Let k be a field. Let V be a vector space over k. Let $\tau : V \times V \to k$ be a bilinear form. We say that τ is non-degenerate if the map $v \mapsto \tau(v, -)$ is injective.

Proposition 5.2.7 Let $\mathbb{F}$ be a field. Let V be a finite dimensional vector space over $\mathbb{F}$. Let $\tau : V \times V \to \mathbb{F}$ be a symmetric bilinear form. Then the following are true. Then the following are equivalent.

- τ is non-degenerate.
- The matrix representing τ over any basis of V is invertible.
- τ is a perfect pairing.
- The map $V \to V^*$ defined by $v \mapsto \tau(v, -)$ is a bijection.

Definition 5.2.8 (Maps Preserving Bilinear Forms)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $\tau : V \times V \to \mathbb{R}$ be a bilinear form. Let $T : V \to V$ be a linear map. We say that T preserves τ if

$$\tau(T(v), T(w)) = \tau(v, w)$$

for all $v, w \in V$.

Lemma 5.2.9
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $\tau : V \times V \to \mathbb{F}$ be a bilinear form. Then the following are true.
- $T, S : V \to V$ preserves τ, then $S \circ T$ preserves τ.
- If $T : V \to V$ preserves τ, then T^{-1} preserves τ.

5.3 Quadratic Forms

Definition 5.3.1 (Quadratic Forms)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $q : V \to \mathbb{F}$ be a function. We say that q is a quadratic form if there exists a symmetric bilinear form $B : V \times V \to \mathbb{F}$ such that

$$q(v) = B(v, v)$$

Under the bijection between bilinear forms on V and $M_n(\mathbb{F})$, the quadratic forms correspond to symmetric matrices. When $\text{char}(\mathbb{F}) \neq 2$, quadratic forms are essentially the same as symmetric bilinear forms in the following way.

Lemma 5.3.2
Let $\mathbb{F}$ be a field such that $\text{char}(\mathbb{F}) \neq 2$. Let V be a vector space over $\mathbb{F}$. Let $q : V \to \mathbb{F}$ be a function. Then the following are equivalent.
- q is a bilinear form.
- $q(\lambda v) = \lambda^2 q(v)$ for all $v \in V$ and all $\lambda \in \mathbb{F}$.
- The map $(u, v) \mapsto \frac{1}{2}(q(u + v) - q(u) - q(v))$ is bilinear.

Lemma 5.3.3 (Polarization Identity)
Let $\mathbb{F}$ be a field such that $\text{char}(\mathbb{F}) \neq 2$. Let V be a vector space over $\mathbb{F}$. Let $q : V \to \mathbb{F}$ be a quadratic form given by $q(v) = B(v, v)$ for some symmetric bilinear form B. Then

$$B(u, v) = \frac{1}{2}(q(u + v) - q(u) - q(v))$$

Theorem 5.3.4 (The Diagonal Form)
Let $\mathbb{F}$ be a field such that $\text{char}(\mathbb{F}) \neq 2$. Let V be a finite dimensional vector space over $\mathbb{F}$. Let $q : V \to \mathbb{F}$ be a quadratic form. Then there exists a basis $b_1, \dots, b_n$ of V such that

$$q\left(\sum_{i=1}^{n} a_i b_i\right) = \sum_{i=1}^{n} c_i a_i^2$$

for some $c_1, \dots, c_n \in \mathbb{F}$.

Definition 5.3.5 (The Quadratic Reduction Algorithm)
Let $\mathbb{F}$ be a field such that $\text{char}(\mathbb{F}) \neq 2$. Let $b_1, \dots, b_n$ be a basis for $\mathbb{F}^n$. Let $q : \mathbb{F}^n \to \mathbb{F}$ be a quadratic form and let $Q = (a_{ij})$ be the corresponding symmetric matrix. The quadratic reduction algorithm is given by the following.

Step 1: Find b_1' such that $q(b_1') \neq 0$. There are a few cases to consider.
- If $a_{11} \neq 0$, then take $b_i' = b_i$ for $1 \leq i \leq n$ and do not change the corresponding symmetric matrix.
- If $a_{11} = 0$ and there is some $1 \leq k \leq n$ such that $a_{kk} \neq 0$, then take $b_1' = b_k$, $b_k' = b_1$ and $b_i' = b_i$ for $i \neq 1, k$. The corresponding symmetric matrix is given by swapping rows r_1 and r_k and then swapping the columns c_1 and c_k of Q.
- If $a_{kk} = 0$ for $1 \leq k \leq n$ but there exists some $1 \leq i, j \leq n$ such that $a_{ij} \neq 0$, then take $b_i' = b_i + b_j$ and $b_k' = b_k$ for $k \neq i$. The corresponding symmetric matrix is given by swapping the row r_i with $r_i + r_j$, then swapping the column c_i with $c_i + c_j$. Then perform

the second bullet point to get $a_{11} \neq 0$.

- If $a_{ij} = 0$ for all $1 \leq i, j \leq n$, then the quadratic form is the zero function.

Step 2: Modify $b_2, \dots, b_n$ so that they are orthogonal to b_1. This is done by setting $b'_k = b_k - \frac{a_{1k}}{a_{11}} b_1$ for $k \neq 1$. The change of basis matrix is given by

$$P = \begin{pmatrix} 1 & -\frac{a_{12}}{a_{11}} & \cdots & -\frac{a_{1n}}{a_{11}} \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}$$

and the corresponding symmetric matrix is given by $P^T Q P$. Step 3: The matrix for the quadratic form now has the form

$$\begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & ? & \cdots & ? \\ \vdots & \vdots & \ddots & \vdots \\ 0 & ? & \cdots & ? \end{pmatrix}$$

We repeat steps 1-3 on the submatrix $(a_{ij})_{2 \leq i,j \leq n}$.

Example 5.3.6 Find a nice basis for the quadratic form $q\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = xy + 3yz - 5xz$.

Proof Using the formula, we construct the matrix of q as

$$A = \begin{pmatrix} 0 & \frac{1}{2} & -\frac{5}{2} \\ \frac{1}{2} & 0 & \frac{3}{2} \\ -\frac{5}{2} & \frac{3}{2} & 0 \end{pmatrix}$$

We start the first change of basis. Notice that $a_{11} = a_{22} = a_{33} = 0$ in the matrix while a_{12} is not. Denote the standard basis by b_1, b_2, b_3. We perform the first basis change by $\{b'_1 = b_1 + b_2, b'_2 = b_2, b'_3 = b_3\}$. This means that the change of basis matrix from new to old is

$$P' = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

To change A into the new matrix A', we simply replace r_1 by $r_1 + r_2$ and replace c_1 by $c_1 + c_2$. This gives

$$A' = \begin{pmatrix} 1 & \frac{1}{2} & -1 \\ \frac{1}{2} & 0 & \frac{3}{2} \\ -1 & \frac{3}{2} & 0 \end{pmatrix}$$

We keep track of changing the basis for clarity. We now have $A' = (P')^T A P'$.
The next step is to set the new basis to $b''_2 = b'_2 - \frac{1}{2} b'_1$ and $b''_3 = b'_3 + b'_1$. This means that the new basis is now $\{b_1 + b_2, \frac{1}{2}(b_2 - b_1), b_1 + b_2 + b_3\}$. The change of basis from this basis back to the old one is now

$$P'' = \begin{pmatrix} 1 & -\frac{1}{2} & 1 \\ 1 & \frac{1}{2} & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Now the new matrix A'' is formed by replacing r_2 with r_2 by $r_2 - \frac{1}{2} r_1$ and r_3 with $r_3 + r_1$. Noticing that A'' must be symmetric, we need to take the new elements and replace the remanining lower traingualr elements so that A'' maintains symmetric. Also observe that the replacement of rows is exactly one changes into the new basis from the previous basis, where $b''_2 = b'_2 - \frac{1}{2} b'_1$ etc. Now we have

$$A'' = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{4} & 2 \\ 0 & 2 & -1 \end{pmatrix}$$

We now have $A'' = (P'')^T AP''$.
Now we perform the next change of basis. We set $b_3''' = b_3'' - \frac{2}{-\frac{1}{4}}b_2'' = b_3'' + 8b_2''$. Now the new basis is $\{b_1 + b_2, \frac{1}{2}(b_2 - b_1), -3b_1 + 5b_2 + b_3\}$. The change of basis matrix is now

$$P''' = \begin{pmatrix} 1 & -\frac{1}{2} & -3 \\ 1 & \frac{1}{2} & 5 \\ 0 & 0 & 1 \end{pmatrix}$$

Similar to the above, we replace r_3 with the same transformation and adjust A''' so that it remains symmetric. Thus now we have

$$A''' = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 15 \end{pmatrix}$$

This means that we are done with $A''' = (P''')^T AP'''$. ■

In general, this result of diagonalization is different from that of similar matrices. One should not be confusing reduction of congruent matrices into diagonal matrices and reduction of similar matrices into JCF as well as reduction of diagonalizable matrices into diagonal matrices.

5.4 Adjoint Maps

Definition 5.4.1 (The Adjoint Map)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $B : V \times V \to \mathbb{F}$ be a bilinear form. Let $T : V \to V$ be a linear map. An adjoint of T is a linear map $T^* : V \to V$ such that

$$B(T(v), w) = B(v, T^*(w))$$

for all $v, w \in V$.

Lemma 5.4.2
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $B : V \times V \to \mathbb{F}$ be a bilinear form. Let $T : V \to V$ be a linear map. Then T^* exists and is unique if and only if B is non-degenerate.

Proposition 5.4.3
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $B : V \times V \to \mathbb{F}$ be a non-degenerate bilinear form. Let $T : V \to V$ be a linear map. Let E be a basis of V. Then we have

$$[T^*]_E^E = [B]_{E,E}^{-1}([T]_E^E)^T([B]_{E,E})$$

Proposition 5.4.4
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $B : V \times V \to \mathbb{F}$ be a non-degenerate bilinear form. Let $T, S : V \to V$ be linear maps. Then the following are true.

- For any $\lambda, \mu \in \mathbb{F}$, we have $(\lambda T + \mu S)^* = \lambda T^* + \mu S^*$.
- We have $(S \circ T)^* = T^* \circ S^*$.

Proposition 5.4.5
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $B : V \times V \to \mathbb{F}$ be a non-degenerate bilinear form. Let $T : V \to V$ be a linear map. Then we have $(T^*)^* = T$.

Definition 5.4.6 (Self and Skew-Adjoints)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $B : V \times V \to \mathbb{F}$ be a non-degenerate bilinear form. Let $T : V \to V$ be a linear map.

- We say that T is self adjoint if $T^* = T$.
- We say that T is skew-adjoint if $T^* = -T$.

5.5 Orthogonality

Definition 5.5.1 (Orthogonal Vectors)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $\tau : V \times V \to \mathbb{F}$ be a symmetric bilinear form. Let $v, w \in V$. We say that v and w are orthogonal if

$$\tau(v, w) = 0$$

Definition 5.5.2 (Orthogonal Complement)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $\tau : V \times V \to \mathbb{F}$ be a symmetric bilinear form. Let $W \subseteq V$ be a subspace of V. Define the orthogonal complement of W to be

$$W^\perp = \{v \in V | \tau(v, w) = 0 \text{ for all } w \in W\}$$

Proposition 5.5.3
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $\tau : V \times V \to \mathbb{F}$ be a symmetric bilinear form. Then the following are equivalent.
- τ is non-degenerate.
- The matrix representing τ over any basis of V is invertible.
- $V^\perp = \{0\}$.

Proposition 5.5.4
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $\tau : V \times V \to \mathbb{F}$ be a symmetric bilinear form. Let W be a subspace of V. Then the following are equivalent.
- $\tau|_{W\times W}$ is non-degenerate on W.
- $W \cap W^\perp = \{0\}$.
- $V = W \oplus W^\perp$.

Proposition 5.5.5
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let $\tau : V \times V \to \mathbb{F}$ be a symmetric bilinear form. Let W be a subspace of V. If $\tau|_{W\times W}$ is non-degenerate on W, then we have

$$W = (W^\perp)^\perp$$

Definition 5.5.6 (Orthgonal Maps)
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let B be a symmetric bilinear form. Let $T : V \to V$ be a linear map. We say that T is orthogonal if

$$B(T(v), T(w)) = B(v, w)$$

for all $v, w \in V$, in other words if T preserves B.

Proposition 5.5.7
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let B be a symmetric bilinear form. Let $T : V \to V$ be a linear map. Then T is orthogonal if and only if

$$[B]_{E,E} = ([T]_E^E)^T [B]_{E,E} [T]_E^E$$

for any basis E of V.

Proposition 5.5.8
Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. Let B be a symmetric non-degenerate bilinear form. Let $T : V \to V$ be a linear map. Then the following are equivalent.
- T is orthogonal.
- The matrix representing T over any basis of V is orthogonal.
- $T^* = T^{-1}$.

6 Inner Products over $\mathbb{R}$

6.1 Norms and Inner Products over $\mathbb{R}$

Definition 6.1.1 (Inner Products over R)
Let V be a vector space over $\mathbb{R}$. An inner product on V is a bilinear form $\langle -,-\rangle : V\times V\to\mathbb{R}$ such that the following are true.
- $\langle -,-\rangle$ is symmetric.
- $\langle -,-\rangle$ is positive definite.

In this case, $(V,\langle -,-\rangle)$ is called an inner product space over $\mathbb{R}$.

Lemma 6.1.2
Let $(V,\langle -,-\rangle)$ be an inner product space over $\mathbb{R}$. Then $\langle -,-\rangle$ is non-degenerate.

Proposition 6.1.3
Let V be a finite dimensional vector space over $\mathbb{R}$. Let $\tau : V\times V\to\mathbb{R}$ be a bilinear form. Then τ is an inner product on V if and only if or any matrix A representing τ over some basis of V, there exists $P\in\mathrm{GL}(\dim(V),\mathbb{R})$ such that $A = P^TP$.

Definition 6.1.4 (The Induced Norm)
Let $(V,\langle -,-\rangle)$ be an inner product space. Define the induced norm of V to be the function $\|\cdot\| : V\to\mathbb{R}$ given by
$$\|x\| = \sqrt{\langle x,x\rangle}$$

Proposition 6.1.5
Let $(V,\langle -,-\rangle)$ be an inner product space. Then the following are true regarding the induced norm.
- Non-negativity: For all $x\in V$, we have $\|x\|\geq 0$.
- Positive-Definite: For all $x\in V$, we have $\|x\| = 0$ if and only if $x = 0$.
- Absolute homogenity: For all $\lambda\in\mathbb{R}$ and all $x\in V$, we have $\|\lambda x\| = |\lambda|\|x\|$.
- For all $x,y\in V$, we have $\|x+y\|\leq\|x\|+\|y\|$.

Lemma 6.1.6 (Polarization Identity)
Let $(V,\langle -,-\rangle)$ be an inner product space over $\mathbb{R}$. Then we have
$$\langle x,y\rangle = \frac{1}{4}\|x+y\|^2 - \frac{1}{4}\|x-y\|^2$$
for all $x,y\in V$.

Proof Simple proof using the fact that $\|x\|^2 = \langle x,x\rangle$. ■

Proposition 6.1.7 (Cauchy-Schwartz Inquality) Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Let $(V,\langle -,-\rangle)$ be an inner product space over $\mathbb{F}$. Let $\|-\|$ be the induced norm from the inner product. For all $x,y\in V$, we have
$$|\langle x,y\rangle|\leq\|x\|\cdot\|y\|$$
with equality if and only if $y = \lambda x$ for some $\lambda\in\mathbb{F}$.

Proof Let $z = x - \frac{|\langle x,y\rangle|}{\|y\|^2}y$. We have $\|z\| \geq 0$.

$$\begin{aligned}\|z\|^2 &= \left\langle x - \frac{|\langle x,y\rangle|}{\|y\|^2}y, x - \frac{|\langle x,y\rangle|}{\|y\|^2}y\right\rangle \\ &= \langle x,x\rangle - 2\left\langle x, \frac{|\langle x,y\rangle|}{\|y\|^2}y\right\rangle + \left\langle \frac{|\langle x,y\rangle|}{\|y\|^2}y, \frac{|\langle x,y\rangle|}{\|y\|^2}y\right\rangle \\ &= \langle x,x\rangle - 2\frac{|\langle x,y\rangle|^2}{\|y\|^2} + \frac{|\langle x,y\rangle|^2}{\|y\|^4}\langle y,y\rangle \\ &= \langle x,x\rangle - 2\frac{|\langle x,y\rangle|^2}{\|y\|^2} + \frac{|\langle x,y\rangle|^2}{\|y\|^2} \\ &= \|x\|^2 - \frac{|\langle x,y\rangle|^2}{\|y\|^2}\end{aligned}$$

Thus we have

$$\begin{aligned}\|x\|^2 &\geq \frac{|\langle x,y\rangle|^2}{\|y\|^2} \\ \|x\| &\geq \frac{|\langle x,y\rangle|}{\|y\|} && \text{(Since they are all positive)} \\ \|x\| \cdot \|y\| &\geq |\langle x,y\rangle|\end{aligned}$$

Note that we have equality if and only if $\|z\| = 0$, meaning $x = \frac{|\langle x,y\rangle|}{\|y\|^2}y$. We are done by taking $\lambda = \frac{\|y\|^2}{|\langle x,y\rangle|}$. ■

6.2 Orthogonality with Inner Products

With the added benefit of positive definiteness, we can produce bases for inner product spaces that are pairwise orthogonal.

Definition 6.2.1 (Orthonormal Basis)
Let $(V, \langle -,-\rangle)$ be an inner product space over $\mathbb{R}$. Let $b_1, \dots, b_n \in V$ be vectors. We say that they form an orthonormal basis of V is the following are true.

- $\{b_1, \dots, b_n\}$ is a basis for V.
- $\|b_i\| = 1$ for $i \in \{1, \dots, n\}$
- The vectors $b_1, \dots, b_n$ are pairwise orthogonal.

Proposition 6.2.2
Let $(V, \langle -,-\rangle)$ be an inner product space over $\mathbb{R}$. Let $T : V \to V$ be a linear map. Then the following are equivalent.

- T is orthogonal.
- The matrix representing T over any basis of V is orthogonal.
- $T^* = T^{-1}$.
- T maps orthonormal bases of V to orthonormal bases.

Example 6.2.3
We have

$$O(n) = \{T : \mathbb{R}^n \to \mathbb{R}^n \mid T \text{ is orthogonal with respect to the standard inner product }\}$$

Proposition 6.2.4
Let $(V, \langle -,-\rangle)$ be an inner product space over $\mathbb{R}$. Let $\| - \|$ be the induced norm. Let $b_1, \dots, b_n$ be

an orthonormal basis for V. Let $v = \sum_{k=1}^n a_k b_k \in V$. Then the following are true.

- $\|v\|^2 = \sum_{k=1}^n |a_k|^2$
- $v = \sum_{k=1}^n \langle v, b_k \rangle b_k$

Proof
We have that

$$\begin{aligned}\|v\|^2 &= \left\langle \sum_{k=1}^n a_k b_k, \sum_{k=1}^n a_k b_k \right\rangle \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j (b_i \cdot b_j) \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \delta_{ij} \\ &= \sum_{k=1}^n |a_k|^2\end{aligned}$$

and we are done.

Applying the inner product with b_i for each $i \in \{1, \dots, n\}$ gives $a_i = \langle v, b_i \rangle$ since $\langle b_i, b_k \rangle = 0$ for any $k \neq i$. Thus if $v = \sum_{k=1}^n a_k b_k$ then $v = \sum_{k=1}^n \langle v, b_k \rangle b_k$ and we are done. ■

Theorem 6.2.5 (Gram-Schmidt Procedure)
Let $(V, \langle -, - \rangle)$ be an inner product space over $\mathbb{F}$. Let $\| - \|$ be the induced norm. Let $v_1, \dots v_m$ be a list of linearly independent vectors of V. Define the following vectors.

- $b_1 = \frac{v_1}{\|v_1\|}$.
- For $i = 2, \dots, m$, define
$$b_i = \frac{v_i - \sum_{k=0}^{i-1} \langle v_i, b_k \rangle b_k}{\|v_i - \sum_{k=0}^{i-1} \langle v_i, b_k \rangle b_k\|}$$

Then $b_1, \dots, b_m$ are orthonormal and has the same span as $v_1, \dots, v_m$.

Corollary 6.2.6
Every finite dimensional inner product space has an orthonormal basis.

Proof
By the Gram-Schmidt procedure, every basis can be transformed into an orthonormal basis. ■

Theorem 6.2.7 (Pythagorean Theorem)
Let $(V, \langle -, - \rangle)$ be an inner product space over $\mathbb{R}$. Let $\| - \|$ be the induced norm. If $u, v \in V$ are orthogonal, then we have
$$\|u + v\|^2 = \|u\|^2 + \|v\|^2$$

Proof
We have that

$$\begin{aligned}\|u + v\|^2 &= \langle u + v, u + v \rangle \\ &= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ &= \langle u, u \rangle + \langle v, v \rangle \\ &= \|u\|^2 + \|v\|^2\end{aligned}$$

■

Theorem 6.2.8 (Orthogonal Decomposition)
Let $(V, \langle -, - \rangle)$ be an inner product space over $\mathbb{R}$. Let $\| - \|$ be the induced norm. Let $u, v \in V$ and $v \neq 0$. Let $w = u - \frac{\langle u,v \rangle}{\|v\|^2} v$. Then the following are true.

- $\langle w, v \rangle = 0$.
- $u = \frac{\langle u,v \rangle}{\|v\|^2} v + w$.

Proof
The fact that $u = cv + w$ is natural so we only have to prove that $\langle w, v \rangle = 0$. We have that

$$\begin{aligned} \langle w, v \rangle &= \left\langle u - \frac{\langle u, v \rangle}{\|v\|^2} v, v \right\rangle \\ &= \langle u, v \rangle - \left\langle \frac{\langle u, v \rangle}{\|v\|^2} v, v \right\rangle \\ &= \langle u, v \rangle - \frac{\langle u, v \rangle}{\|v\|^2} \langle v, v \rangle \\ &= \langle u, v \rangle - \frac{\langle u, v \rangle}{\|v\|^2} \|v\|^2 \\ &= 0 \end{aligned}$$

Thus we are done. ■

6.3 Spectral Theorem over $\mathbb{R}$

Let $(V, \langle -, - \rangle)$ be an inner product space.

- Since $\langle -, - \rangle$ is non-degenerate, we conclude that the adjoint of any linear map exists and is unique.
- Suppose that V is finite dimensional. Since inner products has a suitable basis such that the matrix representing it is the identity map, the matrix of the adjoint of any linear map T is given by

$$[T^*]_{[B]}^{[B]} = \left([T]_{[B]}^{[B]} \right)^T$$

  for some basis B of V.

Proposition 6.3.1 Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be self-adjoint. Then every eigenvalues of T are real.

Proof Suppose that $T(v) = Av$ where A is a representation of T. Suppose that λ is an eigenvalue of T. Then $Av = \lambda v$ for some eigenvector $v \in \mathbb{R}^n$. Then taking complex conjugates give

$$\begin{aligned} \overline{Av} &= \overline{\lambda v} \\ A\overline{v} &= \overline{\lambda}\overline{v} \end{aligned}$$

Taking the transpose of $Av = \lambda v$ gives $v^T A^T = \lambda v^T$ and $v^T A = \lambda v^T$. Multiplying $\overline{v}$ on bothe sides give

$$\begin{aligned} v^T A\overline{v} &= \lambda v^T \overline{v} \\ \overline{\lambda} v^T \overline{v} &= \lambda v^T \overline{v} \end{aligned}$$

But $v^T \overline{v} = v_1 \overline{v_1} + \cdots + v_n \overline{v_n} = |v_1|^2 + \cdots + |v_n|^2$ which is 0 if and only if v is 0. Since eigenvectors are taken to be nonzero, we must have $\lambda = \overline{\lambda}$ and thus λ is real. ■

Theorem 6.3.2 (Spectral Theorem over $\mathbb{R}$)
Let $(V, \langle -, - \rangle)$ be a finite dimensional inner product space over $\mathbb{R}$. Then the following are true.

- Let $T : V \to V$ be a self adjoint linear map. Then V admits an orthonormal basis consisting of eigenvectors of T.
- Let $q : V \to \mathbb{R}$ be a quadratic form, there exists an orthonormal basis of V such that q is of the form $q(x_1, \dots, x_n) = \sum_{i=1}^n c_i x_i^2$ for some $c_1, \dots, c_n \in \mathbb{R}$.

- Let A be a symmetric matrix. Then there exists an orthogonal matrix $O \in O(n, \mathbb{R})$ such that $O^T AO$ is diagonal.

6.4 Decomposition Theorems for Matrices

Theorem 6.4.1 (QR Decomposition) Let A be an $n \times n$ matrix over $\mathbb{R}$. Then we can write $A = QR$ where Q is orthogonal and R is upper triangular.

Proof We split the matrices into two cases. Firstly consider the case where A is invertible. We can treat A as a change of basis matrix from the basis $\{a_1, \dots, a_n\}$ where a_k is the column of A for $k \in \{1, \dots, n\}$. This change of basis matrix takes $\{a_1, \dots, a_n\}$ to $\{e_1, \dots, e_n\}$ which is the standard basis. Apply the Gram-schmidt process to $\{a_1, \dots, a_n\}$ to get $\{b_1, \dots, b_n\}$ which is an orthonormal basis. Let Q be the change of basis matrix from $\{b_1, \dots, b_n\}$ to $\{e_1, \dots, e_n\}$. Let R be the change of basis matrix from $\{a_1, \dots, a_n\}$ to $\{b_1, \dots, b_n\}$. Then clearly $A = QR$. We just have to show that Q is orthogonal and R is upper triangular.
Q being orthonormal is trivial since columns of Q are just $b_1, \dots, b_n$. Using the Gram-schimdt process, we can see that the change of basis from $\{b_1, \dots, b_n\}$ to $\{a_1, \dots, a_n\}$, each b_k is only affected by $a_1, \dots, a_k$ from the old basis. This means that the change of basis matrix must be upper triangular and its inverse must also be upper triangular.
Now we also have the case when A is not invertible. ■

We now give an example of QR decomposition, in conjunction with the Gram-schmidt procedure.

Example 6.4.2 Find the QR decomposition of

$$A = \begin{pmatrix} -1 & 0 & -2 \\ 2 & 0 & -1 \\ 0 & -2 & -2 \end{pmatrix}$$

Proof A quick check shows that A is invertible. Let a_1, a_2, a_3 be the columns of A. We start the Gram-schimdt process by taking the new basis $f_1 = \frac{a_1}{\|a_1\|} = \begin{pmatrix} -\frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \\ 0 \end{pmatrix}$. We also need to keep track on the change of basis matrix. We have that $a_1 = \sqrt{5} f_1$.
For the next step, we find that

$$\begin{aligned} f_2 &= \frac{a_2 - (a_2 \cdot f_1) f_1}{\|a_2 - (a_2 \cdot f_1) f_1\|} \\ &= \frac{a_2}{\|a_2\|} \\ &= \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} \end{aligned}$$

This means that $a_2 = 2f_2$.
Finally, we have that

$$\begin{aligned} f_3 &= \frac{a_3 - (a_3 \cdot f_1) f_1 - (a_3 \cdot f_2) f_2}{\|a_3 - (a_3 \cdot f_1) f_1 - (a_3 \cdot f_2) f_2\|} \\ &= \frac{a_3 - 2f_2}{\|a_3 - 2f_2\|} \\ &= \frac{a_3 - 2f_2}{\sqrt{5}} \\ &= \begin{pmatrix} -\frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \\ 0 \end{pmatrix} \end{aligned}$$

We have that $a_3 = 2f_2 + \sqrt{5}f_3$. Combining everything together, we have that

$$\begin{pmatrix} -1 & 0 & -2 \\ 2 & 0 & -1 \\ 0 & -2 & -2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{\sqrt{5}} & 0 & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & 0 & -\frac{1}{\sqrt{5}} \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} \sqrt{5} & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & \sqrt{5} \end{pmatrix}$$

■

Theorem 6.4.3 (Singular Value Decomposition for Linear Maps) Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear map of rank k. Then there eixsts unique positive numbers $\gamma_1 \geq \gamma_2 \geq \cdots \geq \gamma_k \geq 0$ and orthonormal bases of $\mathbb{R}^n$ and $\mathbb{R}^m$ such that the matrix of T with respect to these bases is

$$\begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}$$

where $D = \text{diag}(\gamma_1, \dots, \gamma_k)$. In fact, the γ_i are exactly the nonzero eigenvalues of T^*T, each one appearing as many times as the dimension of the corresponding eigenspace.

Theorem 6.4.4 (Singular Value Decomposition for Matrices) Let $A_{m\times n}$ be a matrix. There exists unique singular values $\gamma_1 \geq \gamma_2 \geq \dots \gamma_k \geq 0$ where k =rank(A), and orthogonal matrices P, Q such that

$$\begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix} = P^T AQ$$

where $D = \text{diag}(\gamma_1, \dots, \gamma_k)$.

We present an example of singular value decomposition for illustration.

Example 6.4.5 Find the singular value decomposition of the matrix

$$A = \begin{pmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{pmatrix}$$

Proof Step 1: We compute the singular values of A, which is just the squareroot of the eigenvalues of A^TA. Now

$$A^TA = \begin{pmatrix} 80 & 100 & 40 \\ 100 & 170 & 140 \\ 40 & 140 & 200 \end{pmatrix}$$

We have that $c_{A^TA}(x) = x(360 - x)(90 - x)$. This means that the singular values are $\gamma_1 = \sqrt{360} = 6\sqrt{10}$ and $\gamma_2 = \sqrt{90} = 3\sqrt{10}$. Now we want P and Q such that

$$P^TAQ = \begin{pmatrix} 6\sqrt{10} & 0 & 0 \\ 0 & 3\sqrt{10} & 0 \end{pmatrix}$$

Step 2: We find the orthonormal eigenvectors of A^TA so that it forms the matrix Q. This gives

$$Q = \begin{pmatrix} \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{pmatrix}$$

Step 3: We now calculate P by finding the image of the above basis under A, and dividing it with the nonzero singular values. This gives

$$\begin{aligned} P &= \begin{pmatrix} \frac{1}{6\sqrt{10}}Ab_1 & \frac{1}{3\sqrt{10}}Ab_2 \end{pmatrix} \\ &= \begin{pmatrix} \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}} \end{pmatrix} \end{aligned}$$

This means that we are done. ■

7 Symmetric and Skew-Symmetric Bilinear Forms

7.1 Quadratic Forms over $\mathbb{R}$

Theorem 7.1.1 (Sylvester's Law of Inertia)
Let $q : \mathbb{R}^n \to \mathbb{R}$ be a quadratic form. Then the following are true.

- There exists a basis $b_1, \dots, b_n$ of $\mathbb{R}^n$ such that
$$q\left(\sum_{i=1}^n x_i b_i\right) = \sum_{k=1}^t x_k^2 - \sum_{k=1}^u x_k^2$$
where $t + u = \operatorname{rank}(q)$.
- Suppose that $b_1, \dots, b_n$ and $b'_1, \dots, b'_n$ are bass of $\mathbb{R}^n$ such that
$$q\left(\sum_{i=1}^n x_i b_i\right) = \sum_{k=1}^t x_k^2 - \sum_{k=1}^u x_k^2 \quad \text{and} \quad q\left(\sum_{i=1}^n y_i b'_i\right) = \sum_{k=1}^{t'} y_k^2 - \sum_{k=1}^{u'} y_k^2$$
Then we have $t = t'$ and $u = u'$.

Corollary 7.1.2
Let $M \in M_n(\mathbb{R})$ be a symmetric matrix. Then M is congruent to a matrix of the form
$$\begin{pmatrix} I_t & 0 & 0 \\ 0 & -I_u & 0 \\ 0 & 0 & 0 \end{pmatrix}$$
for some $t, u \in \mathbb{N}$.

Definition 7.1.3 (The Signature of a Real Quadratic Form)
Let $q : \mathbb{R}^n \to \mathbb{R}$ be a quadratic form. Let $b_1, \dots, b_n$ be a basis for $\mathbb{R}^n$ such that
$$q\left(\sum_{i=1}^n x_i b_i\right) = \sum_{k=1}^t x_k^2 - \sum_{k=1}^u x_k^2$$
Define the signature of q to be $(t, u) \in \mathbb{N}^2$.

Definition 7.1.4 (The Indefinite Orthogonal Group)
Let $q : \mathbb{R}^n \to \mathbb{R}$ be a quadratic form with signature (t, u). Define the indefinite orthogonal groups by
$$O(t, u) = \{T : \mathbb{R}^n \to \mathbb{R}^n \mid T \text{ preserves } q\}$$

Lemma 7.1.5
The following are true.

- $O(p, q)$ is a group.
- $O(p, q) \cong O(q, p)$.
- $O(p), O(q) \leq O(p, q)$.
- $O(n, 0) \cong O(n) \cong O(0, n)$.

7.2 Symplectic Structures

Definition 7.2.1 (Symplectic Forms)
Let V be a vector space over $\mathbb{R}$. A symplectic form on V is a bilinear map $\omega : V \times V \to \mathbb{R}$ such that the following are true.

- ω is skew-symmetric.
- ω is non-degenerate.

In this case we say that (V,ω) is a symplectic vector space.

Proposition 7.2.2
Let V be a finite dimensional vector space over $\mathbb{R}$. Let ω is a symplectic form on V. Then the following are true.
- Let B be a basis of V. Then $[\omega]_{B,B}$ is a skew-symmetric matrix.
- $\dim(V)$ is even.

Definition 7.2.3 (Symplectic Bases)
Let (V,ω) be a finite dimensional symplectic vector space over $\mathbb{R}$. Let $B = \{e_1,\dots,e_n,f_1,\dots,f_n\}$ be a basis of V. We say that B is symplectic if the following are true.
- $\omega(e_i,e_j) = 0$ for all $1 \leq i,j \leq n$.
- $\omega(f_i,f_j) = 0$ for all $1 \leq i,j \leq n$.
- $\omega(e_i,f_j) = \delta_{ij}$ for all $1 \leq i,j \leq n$.

Definition 7.2.4 (Symplectic Maps)
Let (V,ω) and (W,τ) be symplectic vector spaces over $\mathbb{R}$. Let $T : V \to W$ be a linear map. We say that T is a symplectic map if
$$\tau(T(u),T(v)) = \omega(u,v)$$
for all $u,v \in V$.

Proposition 7.2.5
Let (V,ω) and (W,τ) be symplectic vector spaces over $\mathbb{R}$. Let $T : V \to W$ be a symplectic map. Then the following are true.
- T is injective.
- Suppose that T is an isomorphism. If B is a symplectic basis of V, then $T(B)$ is a symplectic basis of W.

Definition 7.2.6 (Group of Symplectic Maps)
Let (V,ω) be a symplectic vector space over $\mathbb{R}$. Define the group of symplectic maps of V to be the subgroup
$$\text{Sp}(V,J) = \{T : V \to V \mid T \text{ is symplectic }\} \leq \text{GL}(V)$$

Theorem 7.2.7 (Normal Form Theorem for Symplectic Maps)
Let (V,ω) be a symplectic vector space over $\mathbb{R}$. Then the following are true.
- V admits a symplectic basis.
- Let V be a symplectic basis of V. Then we have
$$[\omega]_{B,B} = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$$
- There exists a symplectic isomorphism $T : (V,\omega) \to (\mathbb{R}^{\dim(V)},\omega_{\text{std}})$.
- The above isomorphism induces a group isomorphism
$$\text{Sp}(V,\omega) \cong \text{Sp}(\dim(V),\mathbb{R})$$

8 Sesquilinear Maps

8.1 Conjugate Linear Algebra

Definition 8.1.1 (Conjugate Space)
Let V be a vector space over $\mathbb{C}$. Define the conjugate space of V to be the abelian group

$$\overline{V} = V$$

togerther with the action of $\mathbb{C}$ given by $\lambda \cdot_{\overline{V}} z = \overline{\lambda} \cdot_V z$

Proposition 8.1.2
Let V be a vector space over $\mathbb{C}$. Let B be a basis for V. Then B is also a basis for $\overline{V}$.

Definition 8.1.3 (Conjugate of an Element)
Let V be a vector space over $\mathbb{C}$. Let B be a basis for V. Let $v \in V$ be an element given by $v = \sum_{b \in B} \lambda_b \cdot b$. Define the conjugate of v with respect to B to be

$$\overline{v} = \sum_{b \in B} \overline{\lambda_i} \cdot b$$

Example 8.1.4
Let V be a vector space over $\mathbb{C}$. Then the identity map $\text{id}_V : V \to \overline{V}$ is not a linear map.

Definition 8.1.5 (Conjugate Linear Maps)
Let V, W be vector spaces over $\mathbb{C}$. Let $T : V \to W$ be a function. We say that T is a conjugate linear map if $T : \overline{V} \to W$ is a linear map.

8.2 Matrix Operations over $\mathbb{C}$

Definition 8.2.1 (Conjugate Transpose)
Let $A \in M_{m\times n}(\mathbb{C})$ be a complex matrix. Define the conjugate transpose of A to be

$$A^\dagger = \overline{A}^T$$

where $\overline{A}$ is the matrix of A where you take the conjugate of all elements.

Proposition 8.2.2
Let $A, B \in M_{m\times n}(\mathbb{C})$ be complex matrices. Then we have

$$(\lambda A + \mu B)^\dagger = \overline{\lambda} A^\dagger + \overline{\mu} B^\dagger$$

for any $\lambda, \mu \in \mathbb{C}$.

Proposition 8.2.3
Let $A \in M_{m\times n}(\mathbb{C})$ and $B \in M_{k\times m}(\mathbb{C})$ be matrices. Then we have

$$(BA)^\dagger = A^\dagger B^\dagger$$

Proposition 8.2.4
Let $A \in M_{m\times n}(\mathbb{C})$. Then we have $(A^\dagger)^\dagger$.

Definition 8.2.5 (Hermitian Matrices)
Let $A \in M_n(\mathbb{C})$ be a complex matrix. We say that A is Hermitian if $A^\dagger = A$.

Definition 8.2.6 (Unitary Matrices)
Let $A \in M_n(\mathbb{C})$ be a complex matrix. We say that A is unitary if $A^\dagger = A^{-1}$.

Definition 8.2.7 (Normal Matrices)
Let $A \in M_n(\mathbb{C})$ be a complex matrix. We say that A is normal if $A^\dagger A = AA^\dagger$.

8.3 Sesquilinear Maps

Definition 8.3.1 (Sesquilinear Maps)
Let V, W be vector spaces over $\mathbb{C}$.Let $\tau : V \times W \to \mathbb{C}$ be a function. We say that τ is a sesquilinear map if the following are true.

- $\tau(-, w) : V \to \mathbb{C}$ is a conjugate linear map for all $w \in W$.
- $\tau(v, -) : W \to \mathbb{C}$ is a linear map for all $v \in V$.

Lemma 8.3.2
Let V be a vector space over $\mathbb{C}$. Let $\tau : V \times W \to \mathbb{C}$ be a sesquilinear map. Then τ is a bilinear map $\tau : \overline{V} \times V \to \mathbb{C}$.

Lemma 8.3.3
Let V, W be vector spaces over $\mathbb{C}$. Let $\tau : V \times W \to \mathbb{C}$ be a sesquilinear map. Then there exists functions $B, \omega : V \times W \to \mathbb{R}$ such that the following are true.

- $\tau = B + i\omega$.
- B is a symmetric bilinear form.
- ω is a skew-symmetric bilinear form.

Proposition 8.3.4
Let V, W be vector spaces over $\mathbb{C}$. Let $\tau : V \times W \to \mathbb{C}$ be a sesquilinear map. Let B, C be bases for V and W respectively. Then we have

$$\tau(v, w) = \overline{v}^T [\tau]_{B,C} w$$

for all $v \in V$ and $w \in W$.

Proposition 8.3.5
Let V, W be vector spaces over $\mathbb{C}$. Let $\tau : V \times W \to \mathbb{C}$ be a sesquilinear map. Let E, E' be bases for V and let F, F' be bases for W. Then we have

$$[\tau]_{E',F'} = (C_F^{F'})^\dagger [\tau]_{E,F} C_E^{E'}$$

8.4 Sesquilinear Forms

Definition 8.4.1 (Sesquilinear Forms)
Let V be a vector space over $\mathbb{C}$. A sesquilinear form on V is a sesquilinear map $\tau : V \times V \to \mathbb{C}$.

Definition 8.4.2 (Non-Degenerate Sesquilinear Forms)
Let V be a vector space over $\mathbb{C}$. Let $\tau : V \times V \to \mathbb{C}$ be a sesquilinear form. We say that τ is non-degenerate if the map $v \mapsto \tau(v, -)$ is injective.

Proposition 8.4.3
Let V be a finite dimensional vector space over $\mathbb{C}$. Let $\tau : V \times V \to \mathbb{C}$ be a sesquilinear form. Then the following are equivalent.

- τ is non-degenerate.
- The matrix representing τ over any basis of V is invertible.

- The map $V \to V^*$ defined by $v \mapsto \tau(v, -)$ is a bijection.

Definition 8.4.4 (Hermitian Forms)
Let V be a vector space over $\mathbb{C}$. Let $\tau : V \times V \to \mathbb{C}$ be a sesquilinear form. We say that τ is Hermitian if $\tau(v, w) = \overline{\tau(w, v)}$ for all $v, w \in V$.

Proposition 8.4.5
Let V be a vector space over $\mathbb{C}$. Let $\tau : V \times V \to \mathbb{C}$ be a sesquilinear form. Then τ is Hermitian if and only if the matrix representing τ over any basis is Hermitian.

Lemma 8.4.6
Let V be a vector space over $\mathbb{C}$. Let $\tau : V \times V \to \mathbb{C}$ be a Hermitian form. Let $v \in V$. Then we have

$$\tau(v, v) \in \mathbb{R}$$

8.5 Complex Adjoint Maps

Definition 8.5.1 (The Adjoint Map)
Let V be a vector space over $\mathbb{C}$. Let $B : V \times V \to \mathbb{C}$ be a sesquilinear form. Let $T : V \to V$ be a linear map. An adjoint of T is a conjugate linear map $T^\dagger : V \to V$ such that

$$\tau(T(v), w) = \tau(v, T(w))$$

for all $v, w \in V$.

Lemma 8.5.2
Let V be a vector space over $\mathbb{F}$. Let $\tau : V \times V \to \mathbb{F}$ be a sesquilinear form. Let $T : V \to V$ be a linear map. Then $T^\dagger$ exist and is unique if and only if τ is non-degenerate.

Proposition 8.5.3
Let V be a vector space over $\mathbb{F}$. Let $\tau : V \times V \to \mathbb{F}$ be a non-degenerate sesquilinear form. Let $T : V \to V$ be a linear map. Let E be a basis of V. Then we have

$$[T^\dagger]_E^E = [\tau]_{E,E}([T]_E^E)^\dagger([\tau]_{E,E})^{-1}$$

Proposition 8.5.4
Let V be a vector space over $\mathbb{C}$. Let $\tau : V \times V \to \mathbb{F}$ be a non-degenerate sesquilinear form. Let $T, S : V \to V$ be linear maps. Then the following are true.
- For any $\lambda, \mu \in \mathbb{C}$, we have $(\lambda T + \mu S)^\dagger = \overline{\lambda} T^\dagger + \overline{\mu} S^\dagger$.
- We have $(S \circ T)^\dagger = T^\dagger \circ S^\dagger$.

Proposition 8.5.5
Let V be a vector space over $\mathbb{C}$. Let $\tau : V \times V \to \mathbb{F}$ be a non-degenerate sesquilinear form. Let $T, S : V \to V$ be linear maps. Then we have $(T^\dagger)^\dagger = T$.

Definition 8.5.6 (Self and Skew-Adjoints)
Let V be a vector space over $\mathbb{C}$. Let $\tau : V \times V \to \mathbb{F}$ be a non-degenerate sesquilinear form. Let $T : V \to V$ be a linear map.
- We say that T is self-adjoint if $T^\dagger = T$.
- We say that T is skew-adjoint if $T^\dagger = -T$.

Definition 8.5.7 (Normal Linear Maps)
Let V be a vector space over $\mathbb{C}$. Let $\tau : V \times V \to \mathbb{F}$ be a non-degenerate sesquilinear form. Let

$T : V \to V$ be a linear map. We say that T is normal if $T^\dagger \circ T = T \circ T^\dagger$.

8.6 Unitary Maps

Definition 8.6.1 (Unitary Vectors)
Let V be a vector space over $\mathbb{C}$. Let $\tau : V \times V \to \mathbb{C}$ be a Hermitian form. Let $v, w \in V$. We say that v and w are unitary if $\tau(v, w) = 0$.

Definition 8.6.2 (Unitary Maps)
Let V be a vector space over $\mathbb{C}$. Let $\tau : V \times V \to \mathbb{C}$ be a Hermitian form. Let $T : V \to V$ be a linear map. We say that T is unitary if

$$\tau(T(v), T(w)) = \tau(v, w)$$

for all $v, w \in V$.

Proposition 8.6.3
Let V be a vector space over $\mathbb{C}$. Let $\tau : V \times V \to \mathbb{C}$ be a non-degenerate Hermitian form. Let $T : V \to V$ be a linear map. Then the following are equivalent.
- T is unitary..
- The matrix representing T over any basis of V is unitary.
- $T^\dagger = T^{-1}$.

9 Inner Products over $\mathbb{C}$

9.1 Hermitian Inner Products

Definition 9.1.1 (Hermitian Inner Products)
Let V be a vector space over $\mathbb{C}$. A Hermitian inner product on V is a sesquilinear form $\langle -,-\rangle : V \times V \to \mathbb{R}$ such that the following are true.

- $\langle -,-\rangle$ is Hermitian.
- $\langle -,-\rangle$ is positive definite.

In this case, $(V, \langle -,-\rangle)$ is called a Hermitian inner product space.

Proposition 9.1.2
Let V be a finite dimensional vector space over $\mathbb{C}$. Let $\tau : V \times V \to \mathbb{C}$ be a Hermitian form. Then τ is a Hermitian inner product if and only if or any matrix A representing τ over some basis of V, there exists $P \in \mathrm{GL}(\dim(V), \mathbb{C})$ such that $A = P^\dagger P$.

Definition 9.1.3 (The Induced Norm)
Let $(V, \langle -,-\rangle)$ be a Hermitian inner product space. Define the induced norm of V to be the function $\|\cdot\| : V \to \mathbb{R}$ given by

$$\|x\| = \sqrt{\langle x, x\rangle}$$

Proposition 9.1.4
Let $(V, \langle -,-\rangle)$ be a Hermitian inner product space. Then the following are true regarding the induced norm.

- Non-negativity: For all $x \in V$, we have $\|x\| \geq 0$.
- Positive-Definite: For all $x \in V$, we have $\|x\| = 0$ if and only if $x = 0$.
- Absolute homogenity: For all $\lambda \in \mathbb{R}$ and all $x \in V$, we have $\|\lambda x\| = |\lambda|\|x\|$.
- For all $x, y \in V$, we have $\|x + y\| \leq \|x\| + \|y\|$.

Lemma 9.1.5 (Polarization Identity)
Let $(V, \langle -,-\rangle)$ be a Hermitian inner product space. Then we have

$$\langle u, v\rangle = \frac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2\right)$$

for all $u, v \in V$.

Proposition 9.1.6 (Cauchy-Schwartz Inquality)
Let $(V, \langle -,-\rangle)$ be a Hermitian inner product space. For all $x, y \in V$, we have

$$|\langle x, y\rangle| \leq \|x\| \cdot \|y\|$$

with equality if and only if $y = \lambda x$ for some $\lambda \in \mathbb{F}$.

Proof

Let $z = x - \frac{|\langle x,y\rangle|}{\|y\|^2}y$. We have $\|z\| \geq 0$.

$$\begin{aligned}\|z\|^2 &= \left\langle x - \frac{|\langle x,y\rangle|}{\|y\|^2}y, x - \frac{|\langle x,y\rangle|}{\|y\|^2}y\right\rangle \\ &= \langle x,x\rangle - 2\left\langle x, \frac{|\langle x,y\rangle|}{\|y\|^2}y\right\rangle + \left\langle \frac{|\langle x,y\rangle|}{\|y\|^2}y, \frac{|\langle x,y\rangle|}{\|y\|^2}y\right\rangle \\ &= \langle x,x\rangle - 2\frac{|\langle x,y\rangle|^2}{\|y\|^2} + \frac{|\langle x,y\rangle|^2}{\|y\|^4}\langle y,y\rangle \\ &= \langle x,x\rangle - 2\frac{|\langle x,y\rangle|^2}{\|y\|^2} + \frac{|\langle x,y\rangle|^2}{\|y\|^2} \\ &= \|x\|^2 - \frac{|\langle x,y\rangle|^2}{\|y\|^2}\end{aligned}$$

Thus we have

$$\begin{aligned}\|x\|^2 &\geq \frac{|\langle x,y\rangle|^2}{\|y\|^2} \\ \|x\| &\geq \frac{|\langle x,y\rangle|}{\|y\|} && \text{(Since they are all positive)} \\ \|x\|\cdot\|y\| &\geq |\langle x,y\rangle|\end{aligned}$$

Note that we have equality if and only if $\|z\| = 0$, meaning $x = \frac{|\langle x,y\rangle|}{\|y\|^2}y$. We are done by taking $\lambda = \frac{\|y\|^2}{|\langle x,y\rangle|}$. ■

9.2 Orthonormal Bases

With the added benefit of positive definiteness, we can produce bases for inner product spaces that are pairwise orthogonal.

Definition 9.2.1 (Orthonormal Basis)
Let $(V, \langle -,-\rangle)$ be a Hermitian inner product space. Let $b_1, \dots, b_n \in V$ be vectors. We say that they form a orthonormal basis of V is the following are true.

- $\{b_1, \dots, b_n\}$ is a basis for V.
- $\|b_i\| = 1$ for $i \in \{1, \dots, n\}$
- The vectors $b_1, \dots, b_n$ are pairwise unitary.

Proposition 9.2.2
Let $(V, \langle -,-\rangle)$ be a Hermitian inner product space. Let $T : V \to V$ be a linear map. Then the following are equivalent.

- T is unitary.
- The matrix representing T over any basis of V is unitary.
- $T^\dagger = T^{-1}$.
- T maps orthonormal bases of V to orthonormal bases.

Proposition 9.2.3
Let $(V, \langle -,-\rangle)$ be a Hermitian inner product space. Let $\| - \|$ be the induced norm. Let $b_1, \dots, b_n$ be a orthonormal basis for V. Let $v = \sum_{k=1}^n a_k b_k \in V$. Then the following are true.

- $\|v\|^2 = \sum_{k=1}^n |a_k|^2$
- $v = \sum_{k=1}^n \langle v, b_k\rangle b_k$

Theorem 9.2.4 (Gram-Schmidt Procedure)
Let $(V, \langle -, - \rangle)$ be a Hermitian inner product space. Let $\| - \|$ be the induced norm. Let $v_1, \dots v_m$ be a list of linearly independent vectors of V. Define the following vectors.

- $b_1 = \frac{v_1}{\|v_1\|}$.
- For $i = 2, \dots, m$, define

$$b_i = \frac{v_i - \sum_{k=0}^{i-1} \langle v_i, b_k \rangle b_k}{\|v_i - \sum_{k=0}^{i-1} \langle v_i, b_k \rangle b_k\|}$$

Then $b_1, \dots, b_m$ are orthonormal and has the same span as $v_1, \dots, v_m$.

Corollary 9.2.5
Every finite dimensional Hermitian inner product space has a orthonormal basis.

Proof
By the Gram-Schmidt procedure, every basis can be transformed into an unitary basis. ■

Theorem 9.2.6 (Pythagorean Theorem)
Let $(V, \langle -, - \rangle)$ be a Hermitian inner product space. Let $\| - \|$ be the induced norm. If $u, v \in V$ are unitary, then we have

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2$$

Proof
We have that

$$\begin{aligned}
\|u + v\|^2 &= \langle u + v, u + v \rangle \\
&= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\
&= \langle u, u \rangle + \langle v, v \rangle \\
&= \|u\|^2 + \|v\|^2
\end{aligned}$$

■

Theorem 9.2.7 (Unitary Decomposition)
Let $(V, \langle -, - \rangle)$ be an inner product space over $\mathbb{R}$. Let $\| - \|$ be the induced norm. Let $u, v \in V$ and $v \neq 0$. Let $w = u - \frac{\langle u, v \rangle}{\|v\|^2} v$. Then the following are true.

- $\langle w, v \rangle = 0$.
- $u = \frac{\langle u, v \rangle}{\|v\|^2} v + w$.

Proof
The fact that $u = cv + w$ is natural so we only have to prove that $\langle w, v \rangle = 0$. We have that

$$\begin{aligned}
\langle w, v \rangle &= \left\langle u - \frac{\langle u, v \rangle}{\|v\|^2} v, v \right\rangle \\
&= \langle u, v \rangle - \left\langle \frac{\langle u, v \rangle}{\|v\|^2} v, v \right\rangle \\
&= \langle u, v \rangle - \frac{\langle u, v \rangle}{\|v\|^2} \langle v, v \rangle \\
&= \langle u, v \rangle - \frac{\langle u, v \rangle}{\|v\|^2} \|v\|^2 \\
&= 0
\end{aligned}$$

Thus we are done. ■

9.3 Spectral Theorem over $\mathbb{C}$

Let $(V, \langle -, - \rangle)$ be an inner product space.

- Since $\langle -, - \rangle$ is non-degenerate, we conclude that the adjoint of any linear map exists and is unique.
- Suppose that V is finite dimensional. Since inner products has a suitable basis such that the matrix representing it is the identity map, the matrix of the adjoint of any linear map T is given by

$$[T^*]_{[B]}^{[B]} = \left([T]_{[B]}^{[B]}\right)^T$$

for any basis B of V.

Proposition 9.3.1
Let $(V, \langle -, - \rangle)$ be a Hermitian inner product space. Let $T : V \to V$ be a linear map. If T is self-adjoint, then all eigenvalues of T are real.

Proof Suppose that $T(v) = Av$ where A is a representation of T. Suppose that λ is an eigenvalue of T. Then $Av = \lambda v$ for some eigenvector $v \in \mathbb{R}^n$. Then taking complex conjugates give

$$\begin{aligned}\overline{Av} &= \overline{\lambda v} \\ A\overline{v} &= \overline{\lambda}\overline{v}\end{aligned}$$

Taking the transpose of $Av = \lambda v$ gives $v^T A^T = \lambda v^T$ and $v^T A = \lambda v^T$. Multiplying $\overline{v}$ on bothe sides give

$$\begin{aligned}v^T A\overline{v} &= \lambda v^T \overline{v} \\ \overline{\lambda} v^T \overline{v} &= \lambda v^T \overline{v}\end{aligned}$$

But $v^T\overline{v} = v_1\overline{v_1} + \cdots + v_n\overline{v_n} = |v_1|^2 + \cdots + |v_n|^2$ which is 0 if and only if v is 0. Since eigenvectors are taken to be nonzero, we must have $\lambda = \overline{\lambda}$ and thus λ is real. ■

Theorem 9.3.2 (Spectral Theorem over $\mathbb{C}$)
Let $(V, \langle -, - \rangle)$ be a Hermitian inner product space. Then the following are true.
- Let $T : V \to V$ be a linear map. Then T is normal if and only if V has an orthonormal basis consisting of eigenvectors of T.
- Let $A \in M_n(\mathbb{C})$ be a matrix. Then A is normal if and only if there exists an orthogonal matrix $O \in O(n, \mathbb{C})$ such that $O^T AO$ is diagonal.

9.4 QR Decomposition over $\mathbb{C}$

Theorem 9.4.1 (QR Decomposition over $\mathbb{C}$)
Let $A \in M_n(\mathbb{C})$ be a matrix. Then there exists a unitary matrix $Q \in M_n(\mathbb{C})$ and an upper triangular matrix $R \in M_n(\mathbb{C})$ such that

$$A = QR$$

9.5 Singular Value Decomposition over $\mathbb{C}$

Theorem 9.5.1 (Singular Value Decomposition over $\mathbb{C}$)
Let $A \in M_{m\times n}(\mathbb{C})$ be a matrix of rank k. Then there exists unitary matrices $P \in M_m(\mathbb{C})$ and $Q \in M_n(\mathbb{C})$ and real numbers $\gamma_1 \geq \cdots \geq \gamma_k > 0$ such that

$$P^\dagger AQ = \text{diag}(\gamma_1, \dots, \gamma_k, 0, \dots, 0)$$

Moreover, the $\gamma_1, \dots, \gamma_k$ are the positive square roots of the non-zero eigenvalues of $A^d agger A$.